

Results

v6 analysis compendium for RRRM-2 kidney spaceflight transcriptomics

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Independent research draft for mentor feedback
Authorship/collaboration structure pending

Draft v6 results compendium, May 2026

Abstract

This results-only PDF collects the v6 RRRM-2 kidney transcriptome findings, all figures referenced in the v6 Results section, the core manuscript tables, and the latest analysis-summary tables used to produce the draft. The source runs are the May 19, 2026 production contrast-vector run, the cross-OSDR permutation-sensitivity run, and the 5,000-gene sensitivity run, with WGCNA structural annotation drawn from the remediated WGCNA run referenced by the v6 text.

Source Runs Used for v6

Role	Run ID	Timestamp	Git commit	Parameters/source
Main production contrast run	run_20260519_000547_2500g	2026-05-19T00:05:47.147604	ed36a3559e	max_genes=2500, topk=80, num_seeds=10; data/ results/ run_ 20260519_ 000547_ 2500g/ run_ metadata. json
Cross-OSDR permutation sensitivity run	run_20260519_cosine_perm_null	2026-05-19T19:58:37.484390	ed36a3559e	max_genes=2500, topk=80, num_seeds=10; data/ results/ run_ 20260519_ cosine_ perm_ null/ run_ metadata. json
5,000-gene sensitivity run	run_20260519_5000g_sensitivity_3seed	2026-05-19T00:20:14.323067	ed36a3559e	max_genes=5000, topk=80, num_seeds=3; data/ results/ run_ 20260519_ 5000g_ sensitivity_ 3seed/ run_ metadata. json

Table 1: Run provenance for the analyses that feed the v6 results draft. The contrast-vector manifest for the main run records the tubulointerstitial-state phase with 2,000 bootstrap and 5,000 permutation iterations.

Figure Inventory

Table 2: Every figure referenced in the v6 Results section.

Label	File	Caption summary
fig:mainresult	figures_v4/fig1_main_result_multipanel.pdf	Main result overview: RRRM-2 and OSD-513 remodeling scores, cross-OSDR alignment, and leave-one-pathway-out sensitivity.
fig:alignment	figures_v4/fig4_cross_alignment_leave_one_out.pdf	Cross-OSDR alignment and leave-one-pathway-out sensitivity across the pathway panel.
fig:pathwayscores	figures_v4/fig3_pathway_scores_by_flight.pdf	Axis-defining pathway scores by flight group, emphasizing ECM organization and DCT/NCC-WNK transport.
fig:statespace	figures_v4/fig8_tubulointerstitial_state_space.pdf	Matrix-high / DCT-low tubulointerstitial state space with group-median flight shifts.
fig:stateeffects	figures_v4/fig11_state_component_effects.pdf	Component effects defining the matrix-high / DCT-low state.
fig:deconvoverlay	figures_v4/fig9_rrrm2_deconvolution_overlay.pdf	RRRM-2 deconvolution overlay for the tubulointerstitial state.
fig:renalpanel	figures_v4/fig10_targeted_renal_gene_heatmap.pdf	Targeted renal biology panel in RRRM-2 residualized expression.
fig:larreversal	figures_v4/fig12_lar_reversal_dashboard.pdf	LAR attenuation/divergence and mechanism-context dashboard.
fig:mechheat	figures_v4/fig6_mechanism_score_heatmap.pdf	Mechanism-score associations with the recurrent remodeling score.
fig:osd253	figures_v4/fig5_osd253_strain_points.pdf	OSD-253 remodeling score by strain and control scenario.
fig:genesens	figures_v4/fig7_gene_priority_sensitivity.pdf	Top-k sensitivity for LI-ONESS/node2vec gene prioritization.

Contents

1	Results	5
1.1	RRRM-2 cannot resolve the original age-rewiring contrast at primary resolution .	5
1.2	ISS-T aligns with OSD-513; LAR anti-aligns	6
1.3	The recurrent axis is ECM/fibrosis-positive and DCT/NCC-WNK-negative . . .	7
1.4	A matrix-high / DCT-low tubulointerstitial state emerges across cohort context .	8
1.5	LAR shows attenuation at state-score level and divergence at mechanism-score level	12
1.6	Mechanism scores nominate ECM, innate, adhesion, S1P, and DCT programs . .	14
1.7	OSD-253 points to TLR4/innate association rather than TLR4 requirement . . .	16
1.8	WGCNA provides structural annotation of the remodeling program	16
1.9	LIONESS/node2vec prioritizes candidate local-neighborhood contributors	17
1.10	External aging-axis projection favors remodeling over aging mimicry	17
1.11	Claim hierarchy emerging from the full analysis	18
A	Supplementary Result Tables	23

1 Results

1.1 RRRM-2 cannot resolve the original age-rewiring contrast at primary resolution

RRRM-2 was designed well for a factorial question, but its per-cell sample size is small for estimating high-dimensional aging directions. Bootstrap stability of within-arm GC aging vectors showed that primary module/pathway control-aging directions were unstable in three of four arm-resolution settings (Table 3). LAR module passed, and gene/LIONESS-node summaries passed as secondary resolutions.

Arm	Resolution	Full norm	Median cosine	Lower CI	Required lower	Passed
ISS-T	pathway	0.167	0.787	0.248	0.400	no
ISS-T	module	0.325	0.750	0.056	0.400	no
ISS-T	gene	10.152	0.746	0.545	0.200	yes
ISS-T	LIONESS-node	10.152	0.745	0.549	0.200	yes
LAR	pathway	0.221	0.890	0.345	0.400	no
LAR	module	0.353	0.774	0.493	0.400	yes
LAR	gene	9.687	0.742	0.563	0.200	yes
LAR	LIONESS-node	9.687	0.737	0.556	0.200	yes

Table 3: **Bootstrap angular stability of RRRM-2 control-aging vectors.** Module and pathway are the primary biological resolutions. Gene and LIONESS-node are secondary exploratory resolutions.

This result sets the scale of the manuscript. RRRM-2 provides an arm-stratified flight contrast that can be tested for recurrence in external OSDR cohorts. The secondary stability of gene and LIONESS-node features motivates exploratory local-neighborhood prioritization, while the pathway/module claim rests on cross-cohort flight-vector recurrence.

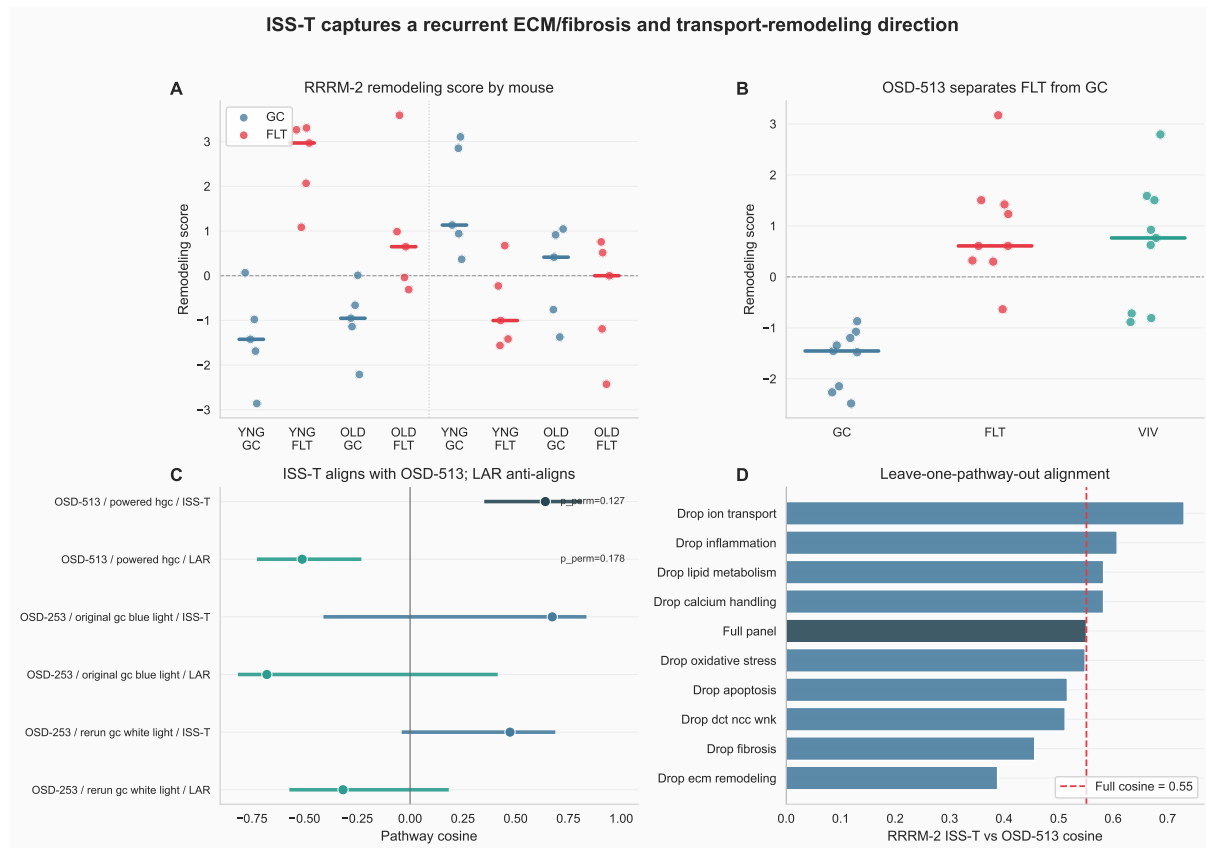


Figure 1: Main result overview. (A) RRRM-2 mouse-level remodeling scores by arm, age, and flight group. Points are individual mice; colored bars are group medians. (B) OSD-513 mouse-level separation on the same remodeling score. (C) Cross-OSDR pathway-vector alignment shows ISS-T alignment with OSD-513 and LAR anti-alignment. (D) Leave-one-pathway-out analysis shows that the alignment is not driven by a single pathway alone.

1.2 ISS-T aligns with OSD-513; LAR anti-aligns

The strongest positive result was arm-specific cross-OSDR alignment at pathway resolution. The RRRM-2 ISS-T flight vector aligned with the OSD-513 flight vector with cosine 0.641 and bootstrap 95% CI 0.351 to 0.814. In the label-permutation sensitivity, the OSD-513 ISS-T cosine was above the null median near zero but did not reach a 0.05 directional permutation threshold ($p_{\text{perm}} = 0.127$). The RRRM-2 LAR vector was anti-aligned with OSD-513, with cosine -0.511 , CI -0.726 to -0.231 , and directional permutation $p = 0.178$ (Table 4; Figure 2). OSD-253 C57BL/6J comparisons had positive ISS-T point estimates under both control definitions but wide confidence intervals crossing zero, so OSD-253 was carried forward as mechanism/context rather than recurrence evidence.

Study	Scenario	Arm	Point	Median	Lower CI	Upper CI	p_{perm}	Boundary	Claimable
OSD-513	powered HGC	ISS-T	0.641	0.625	0.351	0.814	0.127	recurrence	yes
OSD-513	powered HGC	LAR	-0.511	-0.498	-0.726	-0.231	0.178	recurrence	no
OSD-253	original GC blue light	ISS-T	0.674	0.602	-0.411	0.836	0.141	context	no
OSD-253	original GC blue light	LAR	-0.679	-0.608	-0.816	0.415	0.125	context	no
OSD-253	rerun GC white light	ISS-T	0.474	0.470	-0.040	0.688	0.269	context	no
OSD-253	rerun GC white light	LAR	-0.318	-0.309	-0.572	0.183	0.344	context	no

Table 4: **Cross-OSDR pathway-vector alignment.** OSD-513 is interpreted as directional recurrence. OSD-253 is interpreted as context sensitivity because of control-scenario caveats. p_{perm} is the directional external-label permutation sensitivity with the RRRM-2 reference vector held fixed.

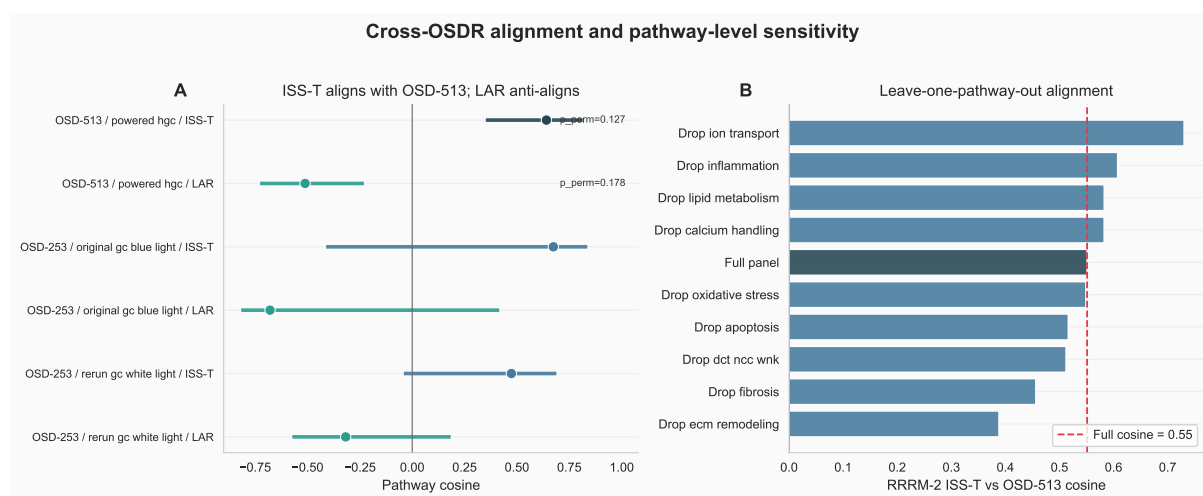


Figure 2: **Cross-OSDR alignment and leave-one-pathway-out sensitivity.** (A) RRRM-2 arm-specific flight vectors compared with external OSDR pathway vectors. ISS-T aligns with OSD-513, whereas LAR anti-aligns; OSD-513 labels show directional external-label permutation p values. (B) Leave-one-pathway-out cosines show that the RRRM-2 ISS-T/OSD-513 alignment is distributed across the pathway panel rather than fully dependent on a single component.

The arm-specific sign pattern is the key biological finding. The terminal-flight arm captures a recurring flight-associated kidney remodeling direction. The return/recovery arm does not recapitulate that direction and is directionally opposite to OSD-513 in the pathway-vector comparison, especially for ECM/fibrosis features. This distinction is consistent with live-return attenuation or context-dependent divergence, while still leaving recovery timing and collection context as linked explanations.

1.3 The recurrent axis is ECM/fibrosis-positive and DCT/NCC-WNK-negative

Pathway-level decomposition identified the features that define the recurring ISS-T/OSD-513 direction. Concordant positive pathways were ECM remodeling, fibrosis, apoptosis, and oxidative stress. DCT/NCC-WNK transport was concordantly negative and therefore received a negative weight. Calcium handling, inflammation, broad ion transport, and lipid metabolism were excluded from the primary axis because they were discordant or mixed. Broad ion transport was retained as sensitivity only. The sample-level pathway scores show the same biological direction: ECM organization increases and DCT/NCC-WNK transport decreases most clearly in the RRRM-2 ISS-T and OSD-513 flight comparisons (Figure 3).

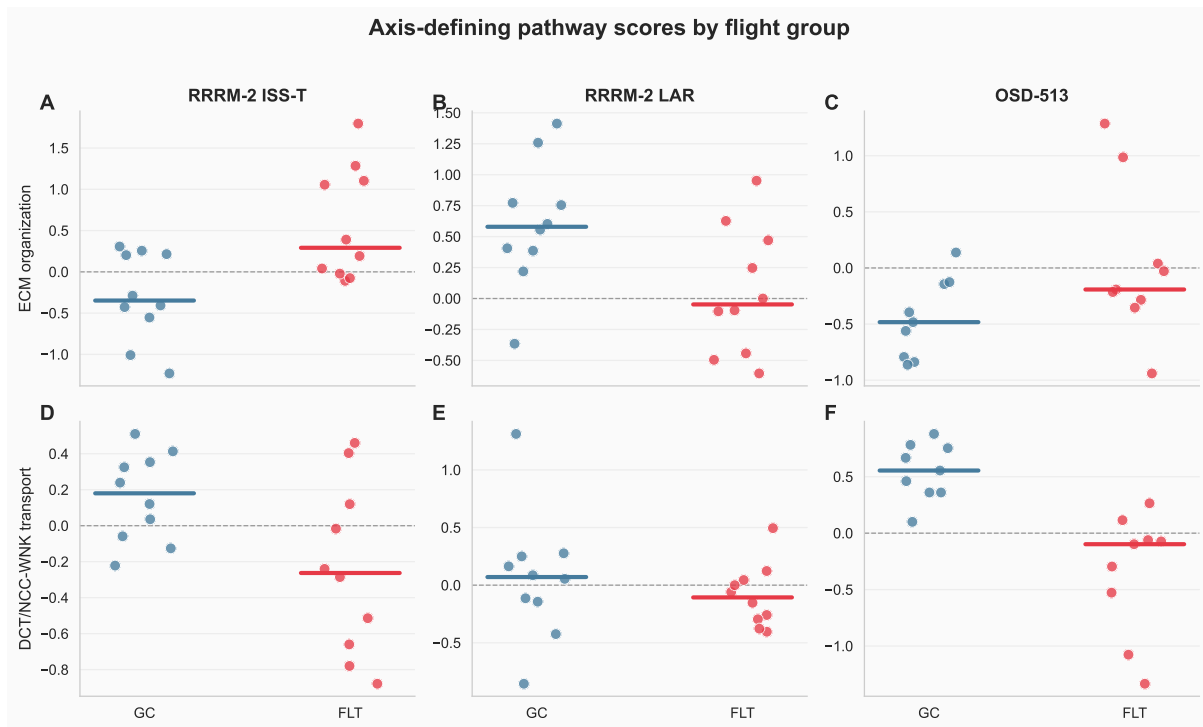


Figure 3: **Axis-defining pathway scores by flight group.** Points are individual mice, and colored bars show medians. ECM organization increases and DCT/NCC-WNK transport decreases in the aligned RRRM-2 ISS-T and OSD-513 flight contexts.

Pathway	RRRM-2 ISS-T effect	OSD-513 effect	Axis weight	Primary axis
ECM remodeling	0.949	0.801	0.622	yes
Fibrosis	0.764	0.652	0.504	yes
DCT/NCC-WNK	-0.398	-0.889	-0.452	yes
Apoptosis	0.674	0.386	0.378	yes
Oxidative stress	0.198	0.104	0.108	yes
Inflammation	0.561	-0.092	0.000	no
Ion transport	0.329	-0.762	0.000	sensitivity
Lipid metabolism	-0.320	0.156	0.000	no
Calcium handling	0.003	-0.550	0.000	no

Table 5: **Pathway construction of the recurrent ISS-T remodeling axis.** Effects are pathway-scale FLT-minus-GC contrasts. The final axis is oriented so ECM/fibrosis is positive and DCT/NCC-WNK suppression is negative.

This axis provides the biological language for the rest of the manuscript: an ECM/fibrosis-positive and DCT/NCC-WNK-negative remodeling score that recurs between RRRM-2 ISS-T and OSD-513.

1.4 A matrix-high / DCT-low tubulointerstitial state emerges across cohort context

The recurrent axis can be read more directly as a two-dimensional tissue-state shift. The matrix component summarizes ECM organization, integrin/cell adhesion, and MMP/ADAM proteolysis. The DCT transport component is kept in its native biological direction, so higher values mean stronger DCT/NCC-WNK transport. The remodeling direction is therefore rightward

and downward in state space, and the scalar `matrix_minus_dct` is defined as matrix component minus DCT transport component (Figure 4).

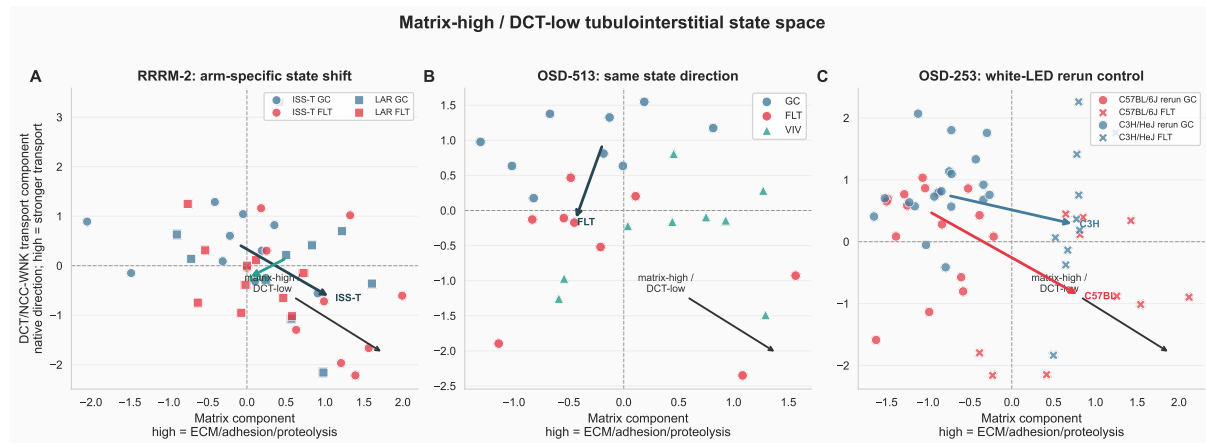


Figure 4: Matrix-high / DCT-low tubulointerstitial state space. The x-axis is the matrix component, where higher values indicate higher ECM/adhesion/proteolysis activity. The y-axis is the DCT/NCC-WNK transport component in native direction, where higher values indicate stronger DCT transport signature. The remodeling direction is rightward and downward. Points are individual mice. Arrows connect group medians from control to flight within the plotted context.

The state-space effects sharpen the biological interpretation (Figure 5). RRRM-2 ISS-T showed a positive matrix shift (FLT-minus-GC effect = 1.25, $q = 0.0158$), a negative DCT transport shift (-1.00 , $q = 0.102$), and a strong positive matrix-minus-DCT state shift (2.25 , $q = 0.0163$). Immune-context and preservation-stress components were also higher in ISS-T flight (1.16 , $q = 0.0012$; 1.06 , $q = 0.0310$). RRRM-2 LAR did not show the same state shift: matrix-minus-DCT was near zero (-0.11 , $q = 0.904$).

OSD-513 was directionally consistent at the scalar state level, with matrix-minus-DCT increasing in flight (1.81 , $q = 0.0310$). This OSD-513 state shift was driven most strongly by DCT/NCC-WNK suppression (-1.57 , $q = 0.0082$), while the matrix component was positive but not significant (0.25 , $q = 0.627$). This distinction matters: OSD-513 supports the same matrix-high/DCT-low direction, but it does not independently reproduce every component with equal strength. The OSD-513 label-permutation caveat remains part of the interpretation.

Under the OSD-253 white-LED rerun control scenario, both strains moved toward the matrix-high/DCT-low state. C57BL/6J showed a strong scalar effect (2.68 , $q = 2.3 \times 10^{-4}$) with matrix increase (1.64 , $q = 8.7 \times 10^{-4}$) and DCT decrease (-1.04 , $q = 0.072$). C3H/HeJ also showed a positive scalar effect (1.82 , $q = 0.0031$) and matrix increase (1.51 , $q = 3.1 \times 10^{-6}$), while its DCT component was only weakly negative (-0.31 , $q = 0.612$). Under the original-GC scenario, the scalar effects were weaker (C57BL/6J 0.79 , $q = 0.229$; C3H/HeJ -0.29 , $q = 0.633$), reinforcing that OSD-253 is sensitive to control definition.

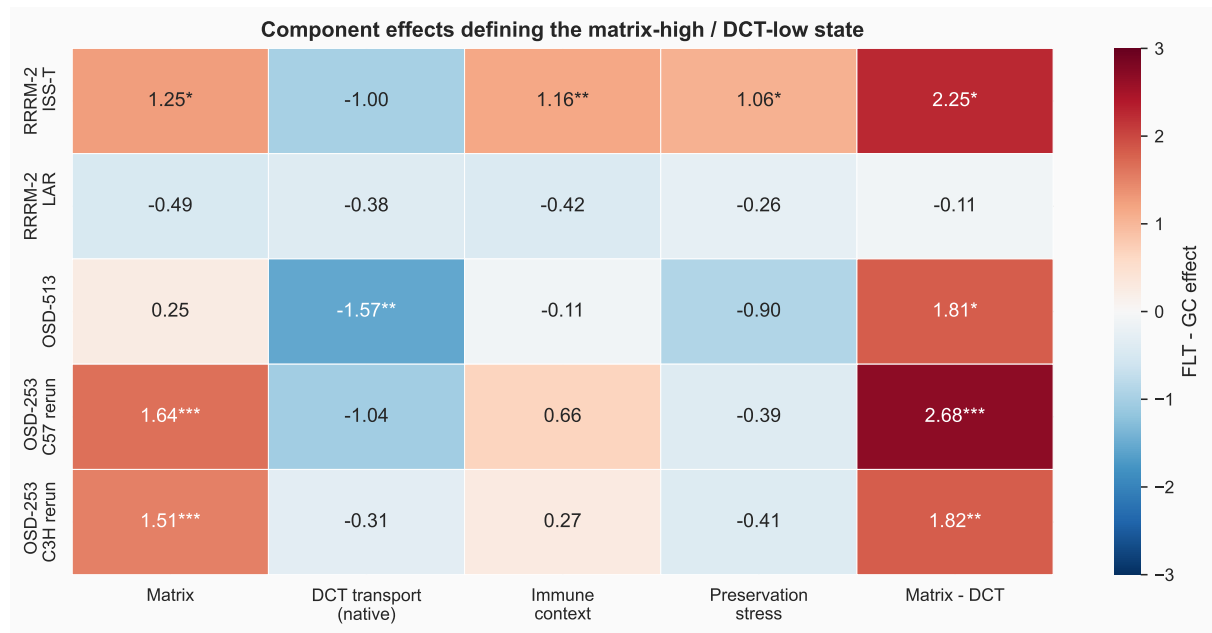


Figure 5: **Component effects defining the matrix-high / DCT-low state.** Values are FLT-minus-GC effects in standardized component units. Asterisks indicate BH-adjusted thresholds within the component-effect table. DCT transport is shown in native direction, so negative values indicate lower DCT/NCC-WNK transport signature in flight.

RRRM-2 deconvolution estimates did not materially track the state components (Figure 6). All 80 RRRM-2 samples joined to the deconvolution table. Across tested state-component/cell-proportion pairs, the largest absolute Spearman correlation was small ($|\rho| = 0.120$ for matrix component versus CLR fibroblast estimate), and all BH-adjusted q values were approximately 0.99. Within the available deconvolution model, the matrix-high/DCT-low state is therefore not simply a readout of estimated DCT or fibroblast fraction.



Figure 6: **RRRM-2 deconvolution overlay for the tubulointerstitial state.** Points are individual RRRM-2 mice. The scalar state score and its components show little correlation with estimated DCT or fibroblast proportions, suggesting that the state-space signal is not explained by these deconvolution estimates alone.

Finally, targeted residualized-expression visualization placed familiar renal genes inside the state interpretation (Figure 7). The panel includes 53 scored genes across ECM/adhesion/proteolysis, TLR/innate/macrophage, DCT/NCC-WNK transport, and tubular/metabolic-priority classes. This is a biological annotation display rather than an additional discovery test, but it is useful for seeing that the state-space result is carried by recognizable renal-remodeling and tubular-transport genes.

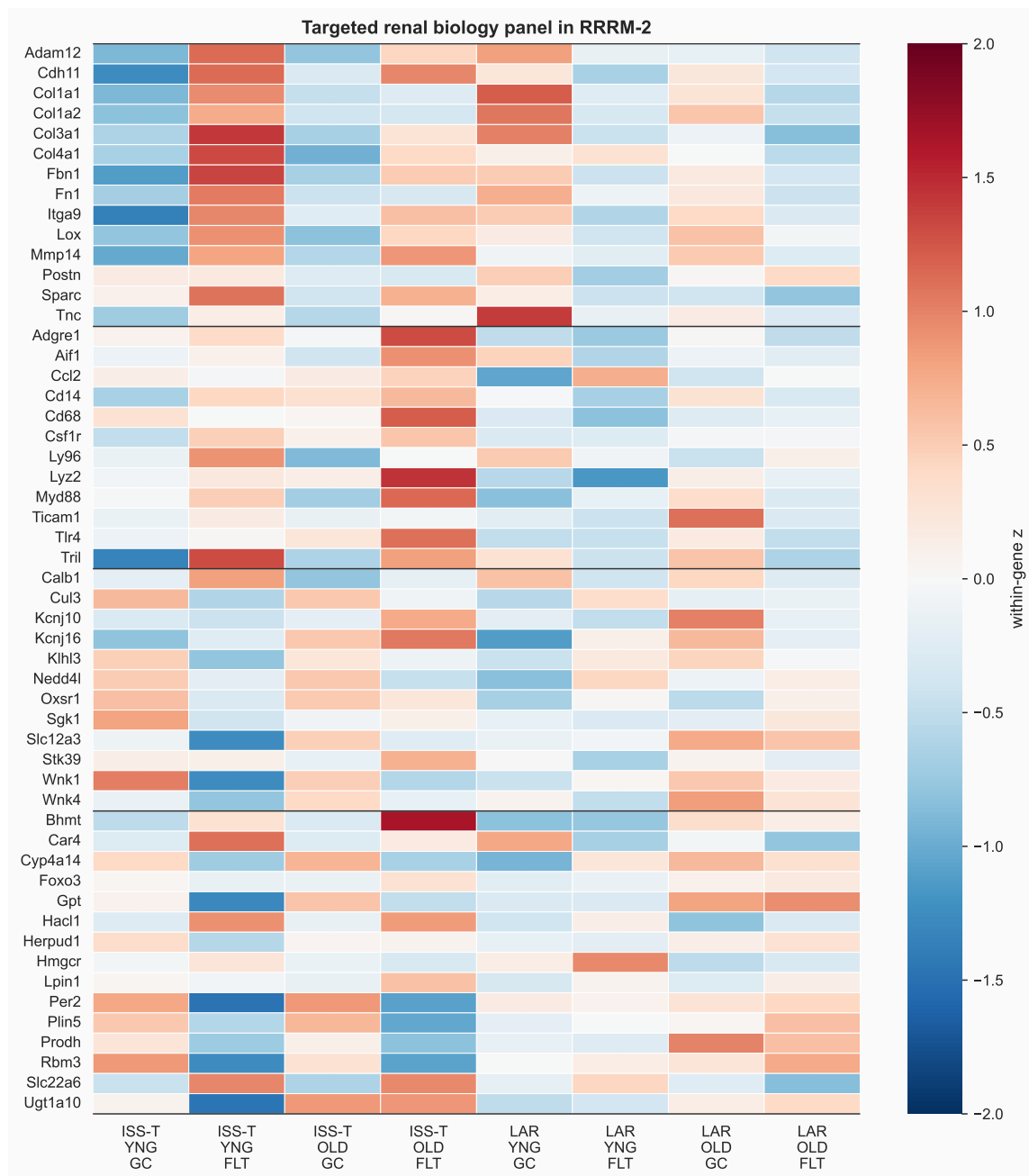


Figure 7: **Targeted renal biology panel in RRRM-2.** Values are group-mean, within-gene z-scores from residualized expression. Rows are grouped by biology class: ECM/adhesion/proteolysis, TLR/innate/macrophage, DCT/NCC-WNK transport, and tubular/metabolic-priority genes.

1.5 LAR shows attenuation at state-score level and divergence at mechanism-score level

The LAR arm was next tested against three competing models: attenuation/recovery toward baseline, true reversal of the ISS-T state, and mechanism switch into a different program. After removing the derived `matrix_minus_dct` scalar from the state-space vector geometry, the pooled LAR state-space vector remained weakly opposite to ISS-T but was unstable (cosine = -0.564 , bootstrap CI -0.940 to $+0.339$, $\beta_{\text{LAR} \rightarrow \text{ISS}} = -0.199$). This supports attenuation rather than a stable state-space reversal. Young LAR had an extreme negative point-estimate cosine (cosine

= -0.982 , directional permutation $p = 0.0030$, BH $q = 0.0450$), but its bootstrap CI remained wide and crossed zero (-0.969 to $+0.559$), so the young state-space direction is not a robust estimate.

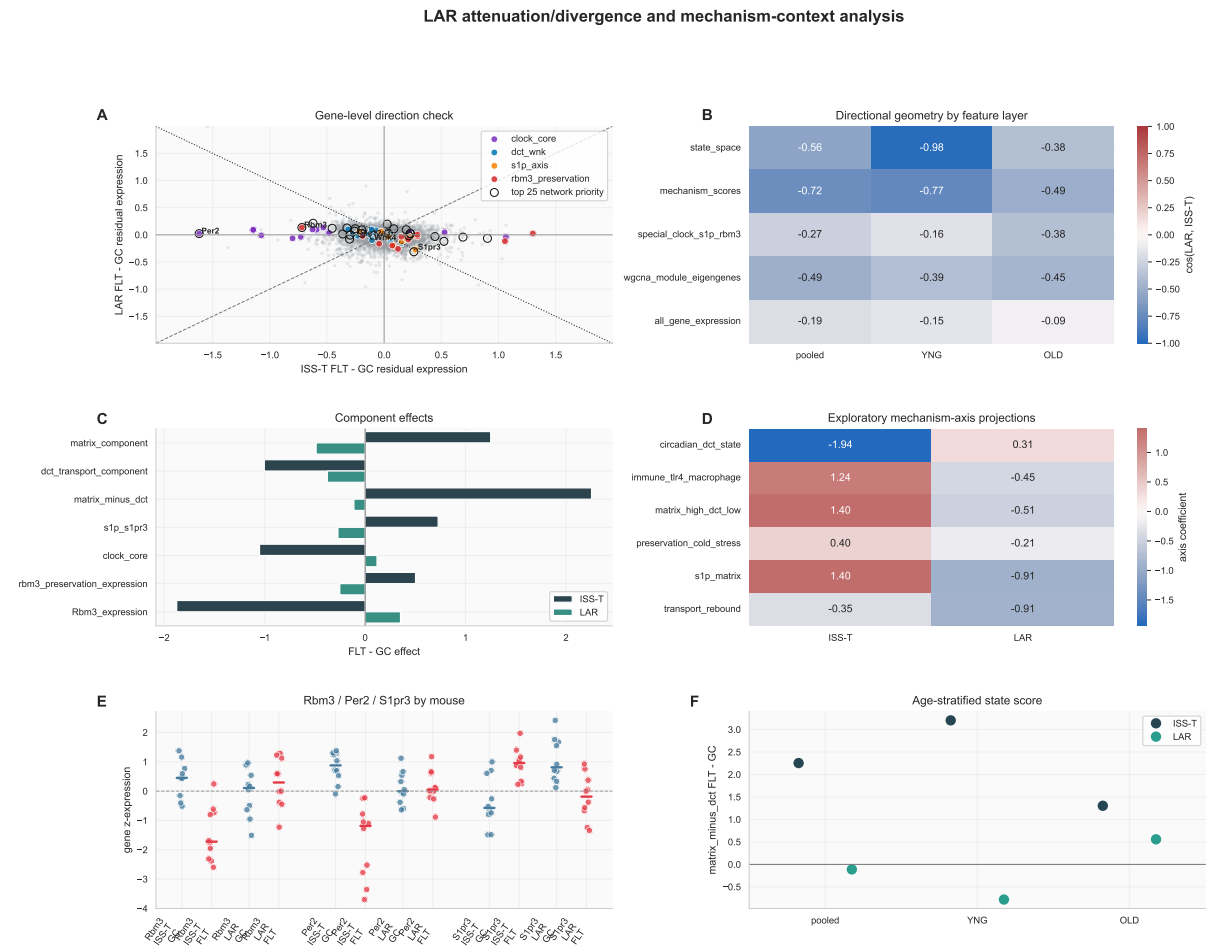


Figure 8: LAR attenuation/divergence and mechanism-context dashboard. Panel A shows RRRM-2 ISS-T and LAR gene-level FLT-minus-GC effects, with curated renal biology classes highlighted. Panel B summarizes cosine alignment across feature sets. Panel C shows arm-specific component effects. Panel D projects ISS-T and LAR effects onto exploratory mechanism-context axes. Panel E shows targeted *Rbm3*, *Per2*, and *S1pr3* expression effects. Panel F shows age-stratified matrix-minus-DCT state effects. The visually prominent young-LAR opposite point estimate is interpreted as hypothesis-generating because the LAR effect itself is not statistically distinguishable from zero.

The scalar state-score effects are most consistent with ISS-T response and LAR attenuation. Young ISS-T moved strongly toward the matrix-high/DCT-low state (matrix-minus-DCT effect = $+3.204$, BH $q = 0.025$), while young LAR had an opposite point estimate (-0.782) that was not statistically distinguishable from zero (Welch $p = 0.361$, BH $q = 0.542$). Old ISS-T showed a weaker positive shift ($+1.304$), and old LAR was weakly positive ($+0.558$). At the pooled level, ISS-T showed a strong state shift ($+2.254$, BH $q = 0.020$), whereas LAR was approximately zero (-0.112 , BH $q = 0.895$). The pooled arm-by-flight interaction for the matrix-minus-DCT state was positive and significant (interaction = 2.366 , bootstrap CI 0.865 to 3.851 , permutation $p = 0.0160$, BH $q = 0.0267$), but this interaction is driven by ISS-T response against an attenuated LAR response, not by a confirmed non-zero LAR reversal.

Component interactions identify which biology differs most between arms. Matrix was the most robust component: ISS-T increased and LAR decreased the matrix component at the point-estimate level (interaction = 1.738, BH $q = 0.0037$), although the LAR matrix effect itself had a bootstrap interval crossing zero. DCT transport was lower in ISS-T and also lower in pooled LAR, and the pooled DCT interaction was not significant (BH $q = 0.346$). Thus, LAR does not show clear DCT recovery.

Mechanism-score vectors provided the strongest evidence that LAR diverges from ISS-T beyond simple state-score attenuation. Pooled mechanism-score vectors showed LAR opposite to ISS-T (cosine = -0.725 , bootstrap CI -0.874 to -0.122 , directional permutation $p = 0.0160$, BH $q = 0.0800$). The clearest component-level divergences involved ECM organization, TLR4/innate signaling, S1P/S1PR3, and *S1pr3* expression. S1P/S1PR3 increased in ISS-T and decreased in LAR (interaction = 0.997, BH $q = 0.0027$), and *S1pr3* expression showed strong arm divergence (ISS-T effect = $+1.276$, LAR effect = -1.269 , interaction = 2.545, BH $q = 0.0020$).

Clock-associated DCT features nominate a second follow-up axis. Core clock score decreased in ISS-T but was slightly positive in LAR (interaction = -1.170 , BH $q = 0.0020$); the Per/Cry subscore showed the same arm divergence (interaction = -2.061 , BH $q = 0.0024$). *Per2* decreased strongly in ISS-T and was approximately unchanged in LAR (interaction = -2.563 , BH $q = 0.0024$). Across RRRM-2 samples, *Per2* expression correlated positively with the DCT transport component ($\rho = 0.449$, BH $q = 7.5 \times 10^{-5}$) and negatively with matrix-minus-DCT ($\rho = -0.547$, BH $q = 7.6 \times 10^{-7}$). Because RRRM-2 does not provide a time-series design or sufficient time-of-day metadata for phase inference, this result is described as a clock-gene-associated transcriptional state, not a measured circadian phase shift.

Exploratory mechanism-axis projections were retained as context, not as confirmatory decomposition. The hand-defined axes explained 37% of the ISS-T mechanism-feature variance and 51% of the LAR variance, leaving substantial residual signal. Their coefficients are therefore useful as biological sketches: matrix/S1P/innate features point toward ISS-T-positive/LAR-negative biology, clock-DCT features suggest ISS-T suppression with LAR attenuation or partial rebound, and transport-rebound features remain negative in both arms. These projections nominate mechanisms to test, but they do not establish three separable programs as a primary result.

At broader feature levels, WGCNA module eigengenes also pointed oppositely (pooled cosine = -0.489 , bootstrap CI -0.763 to -0.080), though permutation support was weaker. All-gene expression was much weaker (cosine = -0.187), arguing against a genome-wide reversal. Node2vec displacement magnitudes were moved to a separate context table because magnitude vectors are non-directional and not comparable to signed effect vectors. Therefore, the LAR result is best framed as pooled attenuation with mechanism-level divergence, plus an underpowered age-stratified observation that may motivate validation.

1.6 Mechanism scores nominate ECM, innate, adhesion, S1P, and DCT programs

Mechanism scoring asked which candidate upstream or contextual programs track the scalar remodeling score. In RRRM-2, the remodeling score correlated strongly with ECM organization ($\rho = 0.644$, $q = 7.4 \times 10^{-10}$), DCT/NCC-WNK transport suppression ($\rho = -0.560$, $q = 3.1 \times 10^{-7}$), TLR4/innate signaling ($\rho = 0.488$, $q = 1.8 \times 10^{-5}$), MMP/ADAM proteolysis ($\rho = 0.478$, $q = 2.6 \times 10^{-5}$), S1P/S1PR3 signaling ($\rho = 0.451$, $q = 8.2 \times 10^{-5}$), and integrin/cell adhesion ($\rho = 0.374$, $q = 1.5 \times 10^{-3}$). OSD-513 reproduced the ECM-positive and DCT-negative relationship: ECM organization correlated with the remodeling score ($\rho = 0.745$, $q = 2.7 \times 10^{-5}$), and DCT/NCC-WNK transport correlated negatively ($\rho = -0.676$, $q = 2.8 \times 10^{-4}$). OSD-253, across all samples, showed strong associations of the remodeling score with TLR4/innate

signaling, ECM organization, macrophage/inflammatory context, DCT/NCC-WNK suppression, and S1P/S1PR3 (Figure 9).

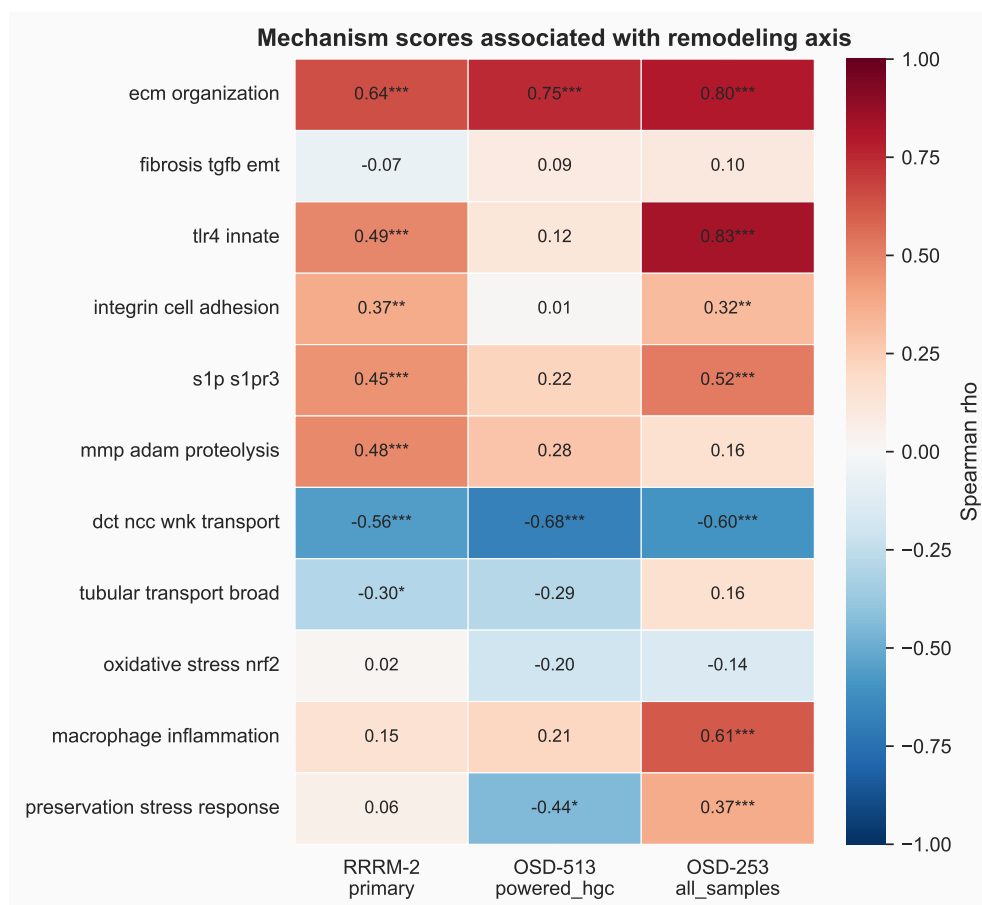


Figure 9: **Mechanism-score associations with the recurrent remodeling score.** Color indicates Spearman correlation between each mechanism score and the scalar remodeling score. Asterisks mark BH-adjusted $q < 0.05$ within the association table. RRRM-2 and OSD-513 both show ECM-positive and DCT/NCC-WNK-negative structure. OSD-253 shows strong TLR4/innate, ECM, macrophage/inflammation, and transport relationships across its broader sample set.

The mechanism pattern is biologically plausible. TLR4 signaling is implicated in renal inflammation and fibrosis, and TLR4-mutant mice have been reported to be protected from chronic kidney disease progression and renal fibrosis [32]. TRIL is a co-receptor associated with TLR3/TLR4 signaling and cytokine production [3, 4, 34]. Cadherin-11 has a recognized role in tissue fibrosis and fibroblast-like cell adhesion [6]. Integrins and MMP/ADAM proteases are core mediators of cell-matrix adhesion and matrix turnover. Sphingosine-1-phosphate signaling, including S1PR3-centered biology, has been linked to kidney injury, inflammation, and fibrotic remodeling in several contexts [27, 28]. The DCT/NCC-WNK direction is also coherent with distal nephron transport regulation, where WNK-SPAK/OSR1-NCC signaling governs sodium-chloride cotransport and potassium-sensitive distal nephron physiology [5, 12].

The mechanism scores place the recurrent remodeling direction in a molecular neighborhood involving ECM deposition, adhesion/proteolysis, innate immune signaling, S1P biology, macrophage/inflammatory context, and tubular transport suppression. This supports a mechanism-prioritization model in which multiple upstream and downstream programs are carried forward for targeted validation.

1.7 OSD-253 points to TLR4/innate association rather than TLR4 requirement

OSD-253 was used as the immediate mechanistic discriminator because C3H/HeJ mice have impaired TLR4 signaling. A strict TLR4-dependent model predicts attenuation of the remodeling score in C3H/HeJ relative to C57BL/6J. The observed pattern supports an association model (Figure 10). Under the original GC blue-light scenario, C57BL/6J remodeling was positive (0.816) and C3H/HeJ remodeling was lower (0.328), with a strain-delta confidence interval crossing zero. Under the white-LED rerun scenario, remodeling was positive in both strains (C57BL/6J 1.379, C3H/HeJ 0.668), again with a strain-delta confidence interval crossing zero. ECM, fibrosis/TGF-beta, integrin/cell adhesion, and MMP/ADAM scores remained positive in C3H/HeJ under the rerun scenario.

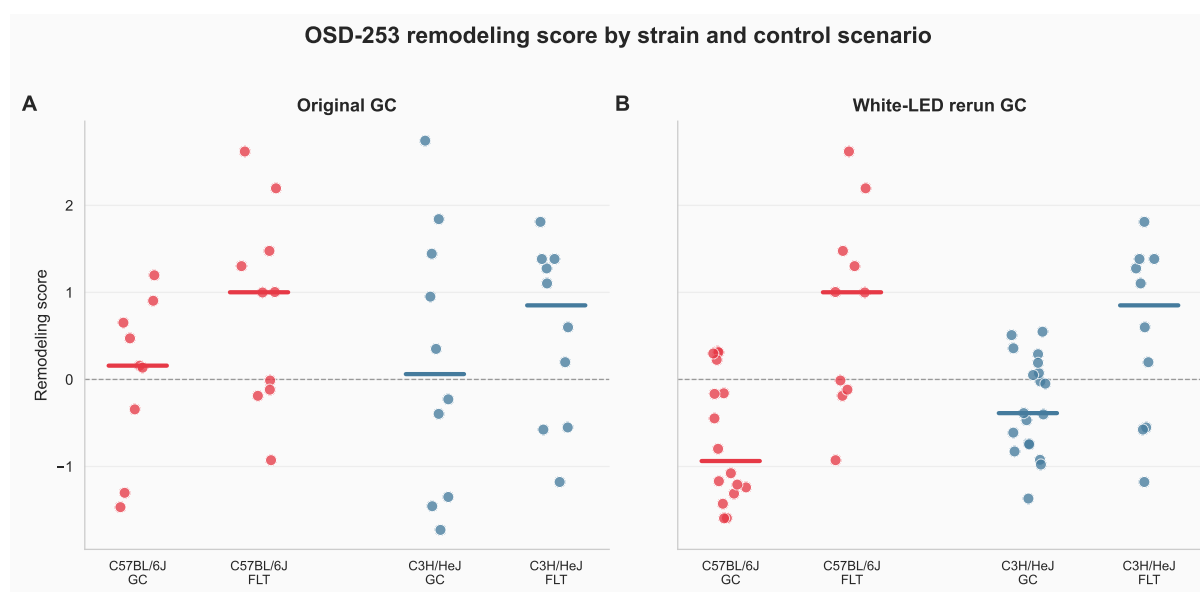


Figure 10: **OSD-253 remodeling score by strain and control scenario.** Points are individual mice, and colored bars show medians. The remodeling score is lower in C3H/HeJ at the point-estimate level, but the program remains positive under the white-LED rerun control scenario.

TLR4/innate signaling tracks the remodeling score strongly, especially in OSD-253, and remains a plausible contextual axis. The strain comparison favors a TLR4-associated but broader remodeling program. Other mechanisms, including TGF-beta/EMT, adhesion, proteolysis, S1P signaling, macrophage context, and preservation stress, remain plausible contributors or covariates.

1.8 WGCNA provides structural annotation of the remodeling program

The WGCNA module layer annotates the remodeling program. A prior WGCNA analysis identified grey60 as a 48-gene ECM/cell-migration module with an ISS-T-specific eigengene pattern and hub structure involving genes such as *Tril*, *Itga9*, *Fbn1*, *S1pr3*, *Cdh11*, *Adam12*, and *Mmp14*. That hub composition is consistent with the mechanism-axis result.

Follow-up overlap analyses place grey60 inside a broader multi-module response. The 11 ISS-T-Young LIONESS factorial edge-rewiring candidates distributed across purple, turquoise, yellow, black, brown, and one off-panel feature. Silent-shifter candidates enriched midnightblue in ISS-T-Young (11/185; $q = 0.0031$) and purple in ISS-T-Old (13/171; $q = 0.030$). Module eigengene

contrasts showed an ISS-T-specific pattern in grey60, royalblue, salmon, and pink. WGCNA therefore supports structural annotation of a multi-module remodeling response, with grey60 as the most biologically interpretable ECM/cell-migration module.

1.9 LIONESS/node2vec prioritizes candidate local-neighborhood contributors

The exploratory LIONESS/node2vec ranking asked which genes show ISS-T-specific local network-neighborhood movement, stronger ISS-T node-strength effects than LAR, and support from OSD-513. In the 2,500-gene production run, the top composite genes included *Rbm3*, *Cyp4a14*, *Bhmt*, *Per2*, *Hmgcr*, *Sim2*, *Hacl1*, *Plin5*, *Gpt*, *Prodh*, and *Ugt1a10*. In the 5,000-gene sensitivity run, high-ranking genes included *Rbm3*, *Bhmt*, *Kank3*, *Ugt1a10*, *Ptpn1*, *Serpib8*, *Cd200*, *Per2*, and *Cyp4a14*.

These genes are candidate local-neighborhood contributors to the pathway-level program. Top-k mechanism enrichment was weak after FDR correction (Figure 11). In the 2,500-gene production run, the strongest top-k findings were broad tubular transport at top 250 (2/4 genes; fold enrichment 5.0; raw $p = 0.052$; $q = 0.384$) and preservation stress at top 250 (3/10; fold enrichment 3.0; raw $p = 0.070$; $q = 0.384$). Ranked-list enrichment suggested broad tubular transport enrichment in the composite score (raw $p = 0.0085$, $q = 0.076$). In the 5,000-gene sensitivity run, the best ranked-list signals were preservation stress, broad tubular transport, and oxidative stress, all with $q = 0.130$ or weaker.

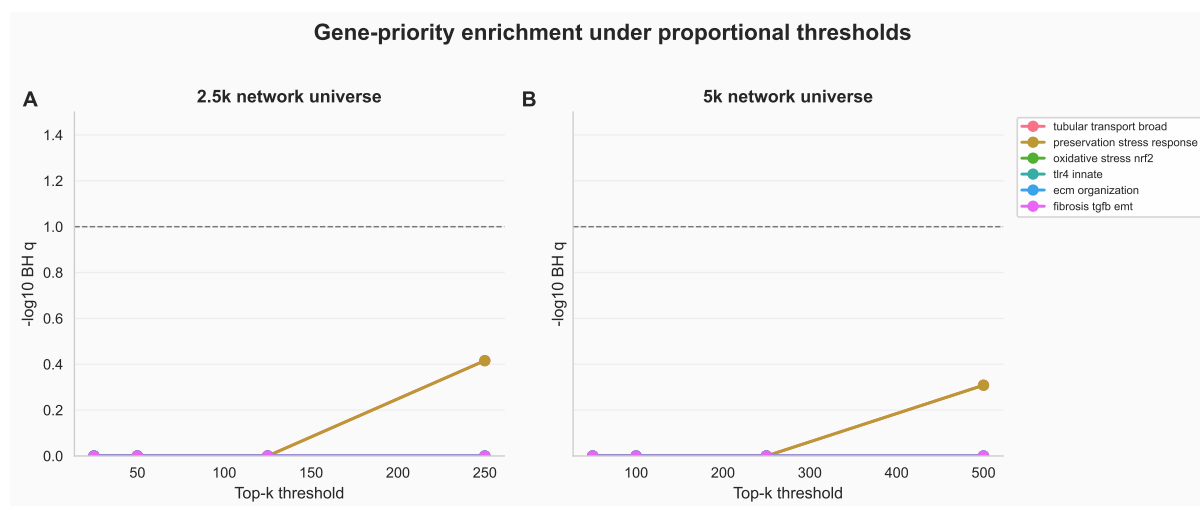


Figure 11: **Top-k sensitivity for LIONESS/node2vec gene prioritization.** The gene-priority layer nominates candidate local-neighborhood contributors. The most consistent weak signals involve broad tubular transport, preservation stress, and oxidative/NRF2 context, while FDR support remains below the pathway/module-scale evidence.

Node2vec and LIONESS add a gene-prioritization layer: they rank genes whose local network context moves with the ISS-T remodeling program. This layer is useful for follow-up target selection and gene-level annotation of the pathway/module result.

1.10 External aging-axis projection favors remodeling over aging mimicry

Projection onto the Tabula Muris Senis kidney aging axis separated the ISS-T remodeling program from simple aging mimicry. ISS-T was essentially orthogonal to the external aging vector ($\beta = -0.0027$, cosine = -0.0187). LAR had a small positive projection ($\beta = 0.0162$,

cosine = 0.175), but the magnitude was small and the residual component remained dominant. The ISS-T signal is therefore best described as recurrent remodeling and transport biology rather than accelerated kidney aging.

1.11 Claim hierarchy emerging from the full analysis

The final claim hierarchy is:

1. RRRM-2 does not resolve the planned within-cohort age-dependent rewiring contrast at primary module/pathway scale.
2. RRRM-2 ISS-T aligns with an independent OSD-513 kidney flight direction; RRRM-2 LAR directionally anti-aligns but does not pass the external-label permutation null.
3. The aligned ISS-T direction is a matrix-high / DCT-low tubulointerstitial state.
4. LAR shows no detectable pooled matrix-minus-DCT flight response, so the primary LAR interpretation is attenuation relative to ISS-T. The young-LAR opposite point estimate is hypothesis-generating because the LAR effect itself is not significant and the state-space bootstrap CI crosses zero.
5. Mechanism-score vectors support LAR divergence from ISS-T. Matrix/S1P/innate, clock-DCT, and tubular-transport projections are retained as follow-up mechanism hypotheses, not as confirmed separable programs.
6. OSD-253 suggests TLR4/innate association, while the C3H/HeJ comparison favors a broader remodeling program rather than strict TLR4 requirement.
7. WGCNA, LIONESS, and node2vec provide structural annotation and exploratory gene prioritization below the cross-cohort pathway/module and mechanism-state results.

Source Table Catalogue

The appendix prints the compact analysis-summary tables used to support the v6 Results text. Large per-sample matrices, bootstrap draws, and raw gene-level tables are indexed here with dimensions and paths so the PDF remains readable while keeping the provenance complete.

Table 6: Tabular outputs from the v6 production, permutation-sensitivity, 5,000-gene sensitivity, and WGCNA-context runs.

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data/results/run_20260519_000547_2500g/contrast_vectors/cross_osdr_recurrence/OSD-253_rerun_gc_white_light_LAR_pathway_bootstrap.tsv	2000	2	indexed
data/results/run_20260519_000547_2500g/contrast_vectors/cross_osdr_recurrence/OSD-513_powered_hgc_ISS-T_pathway_bootstrap.tsv	2000	2	indexed
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A Supplementary Result Tables

The following tables render the compact result summaries from the latest v6 analysis artifacts. Per-sample score matrices and bootstrap-iteration tables are intentionally catalogued rather than printed in full; the printed tables are the human-readable summaries used for manuscript claims and figures.

Table 7: State-space component effects across cohorts and scenarios.

Study	Scenario	Contrast	Arm	Age	Strain	Component	Effect	p	q
RRRM-2	primary	ISS- T_age_pooled_FLT_minus_GC	ISS-T	pooled	NA	matrix_component	1.250	0.002	0.016
RRRM-2	primary	ISS- T_age_pooled_FLT_minus_GC	ISS-T	pooled	NA	dct_transport_component	-1.004	0.035	0.102
RRRM-2	primary	ISS- T_age_pooled_FLT_minus_GC	ISS-T	pooled	NA	immune_context_component	1.158	8.78e-05	0.001
RRRM-2	primary	ISS- T_age_pooled_FLT_minus_GC	ISS-T	pooled	NA	preservation_stress_component	1.062	0.007	0.031
RRRM-2	primary	ISS- T_age_pooled_FLT_minus_GC	ISS-T	pooled	NA	matrix_minus_dct	2.254	0.003	0.016
RRRM-2	primary	ISS- T_YNG_FLT_minus_GC	ISS-T	YNG	NA	matrix_component	1.788	0.016	0.058
RRRM-2	primary	ISS- T_YNG_FLT_minus_GC	ISS-T	YNG	NA	dct_transport_component	-1.416	0.022	0.072
RRRM-2	primary	ISS- T_YNG_FLT_minus_GC	ISS-T	YNG	NA	immune_context_component	0.930	0.002	0.016
RRRM-2	primary	ISS- T_YNG_FLT_minus_GC	ISS-T	YNG	NA	preservation_stress_component	0.689	0.295	0.477
RRRM-2	primary	ISS- T_YNG_FLT_minus_GC	ISS-T	YNG	NA	matrix_minus_dct	3.204	0.005	0.025
RRRM-2	primary	ISS- T_OLD_FLT_minus_GC	ISS-T	OLD	NA	matrix_component	0.712	0.095	0.229
RRRM-2	primary	ISS- T_OLD_FLT_minus_GC	ISS-T	OLD	NA	dct_transport_component	-0.593	0.431	0.586
RRRM-2	primary	ISS- T_OLD_FLT_minus_GC	ISS-T	OLD	NA	immune_context_component	1.385	0.005	0.025
RRRM-2	primary	ISS- T_OLD_FLT_minus_GC	ISS-T	OLD	NA	preservation_stress_component	1.435	0.013	0.052
RRRM-2	primary	ISS- T_OLD_FLT_minus_GC	ISS-T	OLD	NA	matrix_minus_dct	1.304	0.215	0.408
RRRM-2	primary	LAR_age_pooled_FLT_minus_GC	LAR	pooled	NA	matrix_component	-0.488	0.120	0.265
RRRM-2	primary	LAR_age_pooled_FLT_minus_GC	LAR	pooled	NA	dct_transport_component	-0.377	0.461	0.604
RRRM-2	primary	LAR_age_pooled_FLT_minus_GC	LAR	pooled	NA	immune_context_component	-0.424	0.229	0.420
RRRM-2	primary	LAR_age_pooled_FLT_minus_GC	LAR	pooled	NA	preservation_stress_component	-0.265	0.540	0.632
RRRM-2	primary	LAR_age_pooled_FLT_minus_GC	LAR	pooled	NA	matrix_minus_dct	-0.112	0.865	0.904
RRRM-2	primary	LAR_YNG_FLT_minus_GC	LAR	YNG	NA	matrix_component	-0.374	0.435	0.586
RRRM-2	primary	LAR_YNG_FLT_minus_GC	LAR	YNG	NA	dct_transport_component	0.408	0.430	0.586
RRRM-2	primary	LAR_YNG_FLT_minus_GC	LAR	YNG	NA	immune_context_component	-0.177	0.619	0.695
RRRM-2	primary	LAR_YNG_FLT_minus_GC	LAR	YNG	NA	preservation_stress_component	-0.104	0.895	0.912
RRRM-2	primary	LAR_YNG_FLT_minus_GC	LAR	YNG	NA	matrix_minus_dct	-0.782	0.361	0.552
RRRM-2	primary	LAR_OLD_FLT_minus_GC	LAR	OLD	NA	matrix_component	-0.603	0.198	0.389
RRRM-2	primary	LAR_OLD_FLT_minus_GC	LAR	OLD	NA	dct_transport_component	-1.161	0.131	0.277
RRRM-2	primary	LAR_OLD_FLT_minus_GC	LAR	OLD	NA	immune_context_component	-0.672	0.280	0.466

Study	Scenario	Contrast	Arm	Age	Strain	Component	Effect	p	q
RRRM-2	primary	LAR_OLD_FLT_minus_CAR	LAR	OLD	NA	preservation_stress_component	-0.426	0.387	0.575
RRRM-2	primary	LAR_OLD_FLT_minus_CAR	LAR	OLD	NA	matrix_minus_dct	0.558	0.533	0.632
OSD-513	powered_hgc	OSD-513_FLT_minus_GC	NA	pooled	NA	matrix_component	0.248	0.513	0.627
OSD-513	powered_hgc	OSD-513_FLT_minus_GC	NA	pooled	NA	dct_transport_component	-1.566	8.94e-04	0.008
OSD-513	powered_hgc	OSD-513_FLT_minus_GC	NA	pooled	NA	immune_context_component	-0.109	0.754	0.829
OSD-513	powered_hgc	OSD-513_FLT_minus_GC	NA	pooled	NA	preservation_stress_component	-0.900	0.055	0.151
OSD-513	powered_hgc	OSD-513_FLT_minus_GC	NA	pooled	NA	matrix_minus_dct	1.814	0.007	0.031
OSD-253	original_gc_blue_light	OSD-253_original_gc_blue_light_C57BL/6J_FLT_minus_GC	NA	pooled	C57BL/6J	matrix_component	0.749	0.035	0.102
OSD-253	original_gc_blue_light	OSD-253_original_gc_blue_light_C57BL/6J_FLT_minus_GC	NA	pooled	C57BL/6J	dct_transport_component	-0.041	0.921	0.921
OSD-253	original_gc_blue_light	OSD-253_original_gc_blue_light_C57BL/6J_FLT_minus_GC	NA	pooled	C57BL/6J	immune_context_component	0.500	0.248	0.426
OSD-253	original_gc_blue_light	OSD-253_original_gc_blue_light_C57BL/6J_FLT_minus_GC	NA	pooled	C57BL/6J	preservation_stress_component	0.092	0.851	0.904
OSD-253	original_gc_blue_light	OSD-253_original_gc_blue_light_C57BL/6J_FLT_minus_GC	NA	pooled	C57BL/6J	matrix_minus_dct	0.789	0.096	0.229
OSD-253	original_gc_blue_light	OSD-253_original_gc_blue_light_C3H/HeJ_FLT_minus_GC	NA	pooled	C3H/HeJ	matrix_component	0.339	0.089	0.229
OSD-253	original_gc_blue_light	OSD-253_original_gc_blue_light_C3H/HeJ_FLT_minus_GC	NA	pooled	C3H/HeJ	dct_transport_component	0.627	0.195	0.389
OSD-253	original_gc_blue_light	OSD-253_original_gc_blue_light_C3H/HeJ_FLT_minus_GC	NA	pooled	C3H/HeJ	immune_context_component	0.077	0.871	0.904
OSD-253	original_gc_blue_light	OSD-253_original_gc_blue_light_C3H/HeJ_FLT_minus_GC	NA	pooled	C3H/HeJ	preservation_stress_component	0.235	0.513	0.627
OSD-253	original_gc_blue_light	OSD-253_original_gc_blue_light_C3H/HeJ_FLT_minus_GC	NA	pooled	C3H/HeJ	matrix_minus_dct	-0.289	0.552	0.633
OSD-253	rerun_gc_white_light	OSD-253_rerun_gc_white_light_C57BL/6J_FLT_minus_GC	NA	pooled	C57BL/6J	matrix_component	1.641	4.72e-05	8.65e-04
OSD-253	rerun_gc_white_light	OSD-253_rerun_gc_white_light_C57BL/6J_FLT_minus_GC	NA	pooled	C57BL/6J	dct_transport_component	-1.036	0.022	0.072
OSD-253	rerun_gc_white_light	OSD-253_rerun_gc_white_light_C57BL/6J_FLT_minus_GC	NA	pooled	C57BL/6J	immune_context_component	0.658	0.116	0.265
OSD-253	rerun_gc_white_light	OSD-253_rerun_gc_white_light_C57BL/6J_FLT_minus_GC	NA	pooled	C57BL/6J	preservation_stress_component	-0.388	0.437	0.586
OSD-253	rerun_gc_white_light	OSD-253_rerun_gc_white_light_C57BL/6J_FLT_minus_GC	NA	pooled	C57BL/6J	matrix_minus_dct	2.678	8.50e-06	2.34e-04
OSD-253	rerun_gc_white_light	OSD-253_rerun_gc_white_light_C3H/HeJ_FLT_minus_GC	NA	pooled	C3H/HeJ	matrix_component	1.509	5.20e-08	2.86e-06
OSD-253	rerun_gc_white_light	OSD-253_rerun_gc_white_light_C3H/HeJ_FLT_minus_GC	NA	pooled	C3H/HeJ	dct_transport_component	-0.310	0.479	0.612

Study	Scenario	Contrast	Arm	Age	Strain	Component	Effect	p	q
OSD-253	rerun_gc.white.light	OSD- 253_rerun_gc.white.light_C3H/HeJ_FLT_minus_GC	NA	pooled	C3H/HeJ	immune_context_component	0.265	0.335	0.526
OSD-253	rerun_gc.white.light	OSD- 253_rerun_gc.white.light_C3H/HeJ_FLT_minus_GC	NA	pooled	C3H/HeJ	preservation_stress_component	-0.409	0.240	0.426
OSD-253	rerun_gc.white.light	OSD- 253_rerun_gc.white.light_C3H/HeJ_FLT_minus_GC	NA	pooled	C3H/HeJ	matrix_minus_dct	1.819	2.84e-04	0.003

Table 8: RRRM-2 state component correlations with deconvolution estimates.

Study	Scope	Component	Cell estimate	n	rho	p	q
RRRM-2	all_samples	matrix_component	prop_DCT	80	0.014	0.902	0.992
RRRM-2	all_samples	matrix_component	prop_Fibroblast	80	-0.112	0.323	0.992
RRRM-2	all_samples	matrix_component	prop_Immune	80			
RRRM-2	all_samples	matrix_component	prop_Endothelial	80	0.004	0.970	0.992
RRRM-2	all_samples	matrix_component	prop_PT	80	0.043	0.708	0.992
RRRM-2	all_samples	matrix_component	prop_TAL_LOH	80	0.009	0.935	0.992
RRRM-2	all_samples	matrix_component	clr_DCT	80	0.041	0.720	0.992
RRRM-2	all_samples	matrix_component	clr_Fibroblast	80	-0.120	0.288	0.992
RRRM-2	all_samples	matrix_component	clr_Immune	80	0.079	0.484	0.992
RRRM-2	all_samples	dct_transport_component	prop_DCT	80	0.012	0.919	0.992
RRRM-2	all_samples	dct_transport_component	prop_Fibroblast	80	-0.089	0.435	0.992
RRRM-2	all_samples	dct_transport_component	prop_Immune	80			
RRRM-2	all_samples	dct_transport_component	prop_Endothelial	80	-0.073	0.517	0.992
RRRM-2	all_samples	dct_transport_component	prop_PT	80	0.016	0.887	0.992
RRRM-2	all_samples	dct_transport_component	prop_TAL_LOH	80	-0.033	0.771	0.992
RRRM-2	all_samples	dct_transport_component	clr_DCT	80	0.085	0.452	0.992
RRRM-2	all_samples	dct_transport_component	clr_Fibroblast	80	-0.085	0.452	0.992
RRRM-2	all_samples	dct_transport_component	clr_Immune	80	0.054	0.635	0.992
RRRM-2	all_samples	immune_context_component	prop_DCT	80	-0.045	0.689	0.992
RRRM-2	all_samples	immune_context_component	prop_Fibroblast	80	0.022	0.846	0.992
RRRM-2	all_samples	immune_context_component	prop_Immune	80			
RRRM-2	all_samples	immune_context_component	prop_Endothelial	80	-0.058	0.608	0.992
RRRM-2	all_samples	immune_context_component	prop_PT	80	-0.008	0.944	0.992
RRRM-2	all_samples	immune_context_component	prop_TAL_LOH	80	0.007	0.949	0.992
RRRM-2	all_samples	immune_context_component	clr_DCT	80	-0.001	0.992	0.992
RRRM-2	all_samples	immune_context_component	clr_Fibroblast	80	0.041	0.718	0.992
RRRM-2	all_samples	immune_context_component	clr_Immune	80	0.017	0.883	0.992
RRRM-2	all_samples	preservation_stress_component	prop_DCT	80	0.003	0.978	0.992
RRRM-2	all_samples	preservation_stress_component	prop_Fibroblast	80	-0.054	0.636	0.992
RRRM-2	all_samples	preservation_stress_component	prop_Immune	80			
RRRM-2	all_samples	preservation_stress_component	prop_Endothelial	80	-0.031	0.785	0.992
RRRM-2	all_samples	preservation_stress_component	prop_PT	80	0.021	0.853	0.992
RRRM-2	all_samples	preservation_stress_component	prop_TAL_LOH	80	0.008	0.942	0.992
RRRM-2	all_samples	preservation_stress_component	clr_DCT	80	-0.021	0.854	0.992
RRRM-2	all_samples	preservation_stress_component	clr_Fibroblast	80	-0.051	0.650	0.992
RRRM-2	all_samples	preservation_stress_component	clr_Immune	80	0.067	0.555	0.992
RRRM-2	all_samples	matrix_minus_dct	prop_DCT	80	0.018	0.872	0.992
RRRM-2	all_samples	matrix_minus_dct	prop_Fibroblast	80	-0.004	0.974	0.992
RRRM-2	all_samples	matrix_minus_dct	prop_Immune	80			
RRRM-2	all_samples	matrix_minus_dct	prop_Endothelial	80	0.056	0.623	0.992
RRRM-2	all_samples	matrix_minus_dct	prop_PT	80	-0.009	0.937	0.992
RRRM-2	all_samples	matrix_minus_dct	prop_TAL_LOH	80	0.039	0.732	0.992
RRRM-2	all_samples	matrix_minus_dct	clr_DCT	80	-0.039	0.731	0.992

Study	Scope	Component	Cell estimate	n	rho	p	q
RRRM-2	all_samples	matrix_minus_dct	clr_Fibroblast	80	-0.018	0.876	0.992
RRRM-2	all_samples	matrix_minus_dct	clr_Immune	80	-0.007	0.948	0.992

Table 9: Mechanism-score associations with the recurrent remodeling score.

Study	Scenario	Comparison	Mechanism	rho/effect	p	n	q
OSD-253	all_samples	score_vs_remodeling_spearman	ecm_organization	0.801	0	112	0
OSD-253	all_samples	score_vs_remodeling_spearman	fibrosis_tgfb_emt	0.104	0.277	112	0.365
OSD-253	all_samples	score_vs_remodeling_spearman	tlr4_innate	0.832	0	112	0
OSD-253	all_samples	score_vs_remodeling_spearman	integrin_cell_adhesion	0.316	6.78e-04	112	0.001
OSD-253	all_samples	score_vs_remodeling_spearman	s1p_s1pr3	0.517	5.38e-09	112	2.96e-08
OSD-253	all_samples	score_vs_remodeling_spearman	mmp_adam_proteolysis	0.159	0.093	112	0.162
OSD-253	all_samples	score_vs_remodeling_spearman	dct_ncc_wnk_transport	-0.597	3.81e-12	112	3.14e-11
OSD-253	all_samples	score_vs_remodeling_spearman	tubular_transport_broad	0.164	0.085	112	0.155
OSD-253	all_samples	score_vs_remodeling_spearman	oxidative_stress_nrf2	-0.142	0.134	112	0.221
OSD-253	all_samples	score_vs_remodeling_spearman	macrophage_inflammation	0.612	0	112	7.88e-12
OSD-253	all_samples	score_vs_remodeling_spearman	preservation_stress_response	0.371	5.55e-05	112	1.53e-04
OSD-513	powered_hgc	score_vs_remodeling_spearman	ecm_organization	0.745	8.14e-06	27	2.69e-05
OSD-513	powered_hgc	score_vs_remodeling_spearman	fibrosis_tgfb_emt	0.091	0.652	27	0.694
OSD-513	powered_hgc	score_vs_remodeling_spearman	tlr4_innate	0.123	0.540	27	0.636
OSD-513	powered_hgc	score_vs_remodeling_spearman	integrin_cell_adhesion	0.013	0.947	27	0.947
OSD-513	powered_hgc	score_vs_remodeling_spearman	s1p_s1pr3	0.221	0.268	27	0.365
OSD-513	powered_hgc	score_vs_remodeling_spearman	mmp_adam_proteolysis	0.284	0.150	27	0.226
OSD-513	powered_hgc	score_vs_remodeling_spearman	dct_ncc_wnk_transport	-0.676	1.09e-04	27	2.78e-04
OSD-513	powered_hgc	score_vs_remodeling_spearman	tubular_transport_broad	-0.285	0.149	27	0.226
OSD-513	powered_hgc	score_vs_remodeling_spearman	oxidative_stress_nrf2	-0.200	0.318	27	0.389
OSD-513	powered_hgc	score_vs_remodeling_spearman	macrophage_inflammation	0.212	0.289	27	0.367
OSD-513	powered_hgc	score_vs_remodeling_spearman	preservation_stress_response	-0.444	0.020	27	0.039
RRRM-2	primary	score_vs_remodeling_spearman	ecm_organization	0.644	1.12e-10	80	7.37e-10
RRRM-2	primary	score_vs_remodeling_spearman	fibrosis_tgfb_emt	-0.066	0.562	80	0.640
RRRM-2	primary	score_vs_remodeling_spearman	tlr4_innate	0.488	4.48e-06	80	1.85e-05
RRRM-2	primary	score_vs_remodeling_spearman	integrin_cell_adhesion	0.374	6.33e-04	80	0.001
RRRM-2	primary	score_vs_remodeling_spearman	s1p_s1pr3	0.451	2.75e-05	80	8.25e-05
RRRM-2	primary	score_vs_remodeling_spearman	mmp_adam_proteolysis	0.478	7.22e-06	80	2.65e-05
RRRM-2	primary	score_vs_remodeling_spearman	dct_ncc_wnk_transport	-0.560	6.66e-08	80	3.14e-07
RRRM-2	primary	score_vs_remodeling_spearman	tubular_transport_broad	-0.296	0.008	80	0.016
RRRM-2	primary	score_vs_remodeling_spearman	oxidative_stress_nrf2	0.020	0.862	80	0.889
RRRM-2	primary	score_vs_remodeling_spearman	macrophage_inflammation	0.150	0.185	80	0.265
RRRM-2	primary	score_vs_remodeling_spearman	preservation_stress_response	0.062	0.584	80	0.643
RRRM-2	primary	ISS-	ecm_organization	0.859		20	
		T_age_balanced_flight_minus_gc					
RRRM-2	primary	LAR_age_balanced_flight_minus_gc	ecm_organization	-0.545		20	
OSD-513	powered_hgc	flight_minus_gc	ecm_organization	0.485	0.085	18	0.233
RRRM-2	primary	ISS-	fibrosis_tgfb_emt	0.068		20	
		T_age_balanced_flight_minus_gc					
RRRM-2	primary	LAR_age_balanced_flight_minus_gc	fibrosis_tgfb_emt	-0.185		20	
OSD-513	powered_hgc	flight_minus_gc	fibrosis_tgfb_emt	0.082	0.552	18	0.674
RRRM-2	primary	ISS-	tlr4_innate	0.602		20	
		T_age_balanced_flight_minus_gc					

Study	Scenario	Comparison	Mechanism	rho/effect	p	n	q
RRRM-2	primary	LAR_age_balanced_flight_minus_gc	tlr4_innate	-0.223		20	
OSD-513	powered_hgc	flight_minus_gc	tlr4_innate	0.034	0.839	18	0.839
RRRM-2	primary	ISS-	integrin_cell_adhesion	0.445		20	
RRRM-2	primary	T_age_balanced_flight_minus_gc	integrin_cell_adhesion	-0.118		20	
OSD-513	powered_hgc	LAR_age_balanced_flight_minus_gc	integrin_cell_adhesion	-0.120	0.417	18	0.655
RRRM-2	primary	ISS-	slp_slpr3	0.727		20	
RRRM-2	primary	T_age_balanced_flight_minus_gc	slp_slpr3	-0.271		20	
OSD-513	powered_hgc	LAR_age_balanced_flight_minus_gc	slp_slpr3	0.338	0.006	18	0.032
RRRM-2	primary	ISS-	mmp_adam_proteolysis	0.535		20	
RRRM-2	primary	T_age_balanced_flight_minus_gc	mmp_adam_proteolysis	-0.139		20	
OSD-513	powered_hgc	LAR_age_balanced_flight_minus_gc	mmp_adam_proteolysis	0.118	0.291	18	0.583
RRRM-2	primary	ISS-	dct_ncc_wnk_transport	-0.398		20	
RRRM-2	primary	T_age_balanced_flight_minus_gc	dct_ncc_wnk_transport	-0.149		20	
OSD-513	powered_hgc	LAR_age_balanced_flight_minus_gc	dct_ncc_wnk_transport	-0.889	8.94e-04	18	0.010
RRRM-2	primary	ISS-	tubular_transport_broad	-0.267		20	
RRRM-2	primary	T_age_balanced_flight_minus_gc	tubular_transport_broad	-0.167		20	
OSD-513	powered_hgc	LAR_age_balanced_flight_minus_gc	tubular_transport_broad	-0.190	0.522	18	0.674
RRRM-2	primary	ISS-	oxidative_stress_nrf2	-0.047		20	
RRRM-2	primary	T_age_balanced_flight_minus_gc	oxidative_stress_nrf2	0.271		20	
OSD-513	powered_hgc	LAR_age_balanced_flight_minus_gc	oxidative_stress_nrf2	-0.044	0.810	18	0.839
RRRM-2	primary	ISS-	macrophage_inflammation	0.380		20	
RRRM-2	primary	T_age_balanced_flight_minus_gc	macrophage_inflammation	-0.135		20	
OSD-513	powered_hgc	LAR_age_balanced_flight_minus_gc	macrophage_inflammation	-0.214	0.318	18	0.583
RRRM-2	primary	ISS-	preservation_stress_response	0.594		20	
RRRM-2	primary	T_age_balanced_flight_minus_gc	preservation_stress_response	-0.148		20	
OSD-513	powered_hgc	LAR_age_balanced_flight_minus_gc	preservation_stress_response	-0.537	0.055	18	0.202

Table 10: OSD-253 TLR4/innate discriminator summary.

Scenario	Score	C57 effect	C3H effect	Delta	C57 q	C3H q	Delta q	Interpretation
original_gc_blue_light	remodeling_score	0.816	0.328	-0.488	0.279	0.757	0.748	inconclusive
original_gc_blue_light	tlr4_innate	0.184	0.231	0.048	0.460	0.521	0.870	tlr4_not_required
original_gc_blue_light	fibrosis_tgfb_emt	0.190	-0.038	-0.227	0.279	0.757	0.550	inconclusive
original_gc_blue_light	ecm_organization	0.507	0.398	-0.110	0.279	0.308	0.870	tlr4_not_required
original_gc_blue_light	integrin_cell_adhesion	0.336	0.187	-0.148	0.182	0.308	0.748	inconclusive
original_gc_blue_light	mmp_adam_proteolysis	0.316	-0.051	-0.366	0.279	0.757	0.550	inconclusive
original_gc_blue_light	dct_ncc_wnk_transport	-0.030	0.214	0.244	0.791	0.463	0.550	inconclusive
rerun_gc_white_light	remodeling_score	1.379	0.668	-0.710	0.018	0.132	0.925	inconclusive
rerun_gc_white_light	tlr4_innate	0.229	0.149	-0.080	0.220	0.363	0.925	inconclusive
rerun_gc_white_light	fibrosis_tgfb_emt	0.714	0.547	-0.167	0.003	8.75e-04	0.925	tlr4_not_required
rerun_gc_white_light	ecm_organization	0.798	0.723	-0.075	0.010	8.75e-04	0.925	tlr4_not_required
rerun_gc_white_light	integrin_cell_adhesion	0.869	0.836	-0.033	0.003	8.75e-04	0.925	tlr4_not_required
rerun_gc_white_light	mmp_adam_proteolysis	0.779	0.687	-0.093	0.003	8.75e-04	0.925	tlr4_not_required
rerun_gc_white_light	dct_ncc_wnk_transport	-0.356	-0.106	0.250	0.031	0.506	0.925	inconclusive

Table 11: LAR attenuation/divergence vector summaries.

Feature set	Age	n	cos	CI low	CI high	beta	perm p	q	Interpretation
state_space	pooled	4	-0.564	-0.940	0.339	-0.199	0.177	0.333	model_A_attenuation_candidate
state_space	YNG	4	-0.982	-0.969	0.559	-0.227	0.003	0.045	model_A_attenuation_candidate
state_space	OLD	4	-0.381	-0.882	0.421	-0.265	0.293	0.343	model_C_partial_reversal_or_switch
mechanism_scores	pooled	11	-0.725	-0.874	-0.122	-0.347	0.016	0.080	model_B_reversal_candidate
mechanism_scores	YNG	11	-0.766	-0.871	-0.090	-0.382	0.011	0.080	model_B_reversal_candidate
mechanism_scores	OLD	11	-0.495	-0.797	0.264	-0.310	0.099	0.249	model_C_partial_reversal_or_switch
special_clock_slp_rbm3	pooled	12	-0.273	-0.763	0.244	-0.092	0.297	0.343	model_A_attenuation_candidate
special_clock_slp_rbm3	YNG	12	-0.161	-0.753	0.387	-0.054	0.380	0.380	model_A_attenuation_candidate
special_clock_slp_rbm3	OLD	12	-0.379	-0.865	0.277	-0.151	0.263	0.343	model_A_attenuation_candidate
all_gene_expression	pooled	5000	-0.187	-0.451	0.169	-0.124	0.219	0.343	mixed_or_inconclusive
all_gene_expression	YNG	5000	-0.145	-0.415	0.240	-0.113	0.281	0.343	mixed_or_inconclusive
all_gene_expression	OLD	5000	-0.086	-0.390	0.285	-0.052	0.347	0.372	mixed_or_inconclusive
wgcna_module_eigengenes	pooled	21	-0.489	-0.763	-0.080	-0.280	0.074	0.249	model_B_reversal_candidate
wgcna_module_eigengenes	YNG	21	-0.387	-0.723	0.124	-0.230	0.156	0.333	model_A_attenuation_candidate
wgcna_module_eigengenes	OLD	21	-0.452	-0.732	0.063	-0.256	0.089	0.249	model_C_partial_reversal_or_switch
lioness_node_strength_summary	pooled	2500	-0.183			-0.115			exploratory_precomputed_network_summary

Table 12: Complete LAR component-effect table used by the dashboard and text.

Family	Feature	Arm	Age	FLT n	GC n	Effect	p	q
state_space	matrix_component	ISS-T	pooled	10	10	1.250	0.002	0.020
state_space	matrix_component	LAR	pooled	10	10	-0.488	0.120	0.278
state_space	matrix_component	ISS-T	YNG	5	5	1.788	0.016	0.053
state_space	matrix_component	LAR	YNG	5	5	-0.374	0.435	0.543
state_space	matrix_component	ISS-T	OLD	5	5	0.712	0.095	0.237
state_space	matrix_component	LAR	OLD	5	5	-0.603	0.198	0.396
state_space	dct_transport_component	ISS-T	pooled	10	10	-1.004	0.035	0.094
state_space	dct_transport_component	LAR	pooled	10	10	-0.377	0.461	0.553
state_space	dct_transport_component	ISS-T	YNG	5	5	-1.416	0.022	0.067
state_space	dct_transport_component	LAR	YNG	5	5	0.408	0.430	0.543
state_space	dct_transport_component	ISS-T	OLD	5	5	-0.593	0.431	0.543
state_space	dct_transport_component	LAR	OLD	5	5	-1.161	0.131	0.281
state_space	immune_context_component	ISS-T	pooled	10	10	1.158	8.78e-05	0.003
state_space	immune_context_component	LAR	pooled	10	10	-0.424	0.229	0.405
state_space	immune_context_component	ISS-T	YNG	5	5	0.930	0.002	0.020
state_space	immune_context_component	LAR	YNG	5	5	-0.177	0.619	0.663
state_space	immune_context_component	ISS-T	OLD	5	5	1.385	0.005	0.025
state_space	immune_context_component	LAR	OLD	5	5	-0.672	0.280	0.465
state_space	preservation_stress_component	ISS-T	pooled	10	10	1.062	0.007	0.031
state_space	preservation_stress_component	LAR	pooled	10	10	-0.265	0.540	0.600
state_space	preservation_stress_component	ISS-T	YNG	5	5	0.689	0.295	0.465
state_space	preservation_stress_component	LAR	YNG	5	5	-0.104	0.895	0.895
state_space	preservation_stress_component	ISS-T	OLD	5	5	1.435	0.013	0.050
state_space	preservation_stress_component	LAR	OLD	5	5	-0.426	0.387	0.543
state_space	matrix_minus_dct	ISS-T	pooled	10	10	2.254	0.003	0.020
state_space	matrix_minus_dct	LAR	pooled	10	10	-0.112	0.865	0.895
state_space	matrix_minus_dct	ISS-T	YNG	5	5	3.204	0.005	0.025
state_space	matrix_minus_dct	LAR	YNG	5	5	-0.782	0.361	0.542
state_space	matrix_minus_dct	ISS-T	OLD	5	5	1.304	0.215	0.403
state_space	matrix_minus_dct	LAR	OLD	5	5	0.558	0.533	0.600
mechanism_scores	ecm_organization	ISS-T	pooled	10	10	0.859	0.006	0.077
mechanism_scores	ecm_organization	LAR	pooled	10	10	-0.545	0.028	0.121
mechanism_scores	ecm_organization	ISS-T	YNG	5	5	1.248	0.024	0.115
mechanism_scores	ecm_organization	LAR	YNG	5	5	-0.833	0.029	0.121
mechanism_scores	ecm_organization	ISS-T	OLD	5	5	0.470	0.114	0.297
mechanism_scores	ecm_organization	LAR	OLD	5	5	-0.257	0.452	0.606
mechanism_scores	fibrosis_tgfb_emt	ISS-T	pooled	10	10	0.068	0.699	0.796
mechanism_scores	fibrosis_tgfb_emt	LAR	pooled	10	10	-0.185	0.117	0.297
mechanism_scores	fibrosis_tgfb_emt	ISS-T	YNG	5	5	-0.021	0.934	0.949
mechanism_scores	fibrosis_tgfb_emt	LAR	YNG	5	5	-0.146	0.454	0.606
mechanism_scores	fibrosis_tgfb_emt	ISS-T	OLD	5	5	0.156	0.538	0.652
mechanism_scores	fibrosis_tgfb_emt	LAR	OLD	5	5	-0.223	0.063	0.207
mechanism_scores	tlr4_innate	ISS-T	pooled	10	10	0.602	1.23e-05	8.13e-04

Family	Feature	Arm	Age	FLT n	GC n	Effect	p	q
mechanism_scores	tlr4_innate	LAR	pooled	10	10	-0.223	0.066	0.208
mechanism_scores	tlr4_innate	ISS-T	YNG	5	5	0.522	4.00e-04	0.009
mechanism_scores	tlr4_innate	LAR	YNG	5	5	-0.073	0.544	0.652
mechanism_scores	tlr4_innate	ISS-T	OLD	5	5	0.682	0.008	0.077
mechanism_scores	tlr4_innate	LAR	OLD	5	5	-0.372	0.099	0.284
mechanism_scores	integrin_cell_adhesion	ISS-T	pooled	10	10	0.445	0.012	0.086
mechanism_scores	integrin_cell_adhesion	LAR	pooled	10	10	-0.118	0.456	0.606
mechanism_scores	integrin_cell_adhesion	ISS-T	YNG	5	5	0.622	0.034	0.127
mechanism_scores	integrin_cell_adhesion	LAR	YNG	5	5	0.063	0.812	0.864
mechanism_scores	integrin_cell_adhesion	ISS-T	OLD	5	5	0.268	0.172	0.343
mechanism_scores	integrin_cell_adhesion	LAR	OLD	5	5	-0.299	0.147	0.313
mechanism_scores	s1p_s1pr3	ISS-T	pooled	10	10	0.727	3.89e-04	0.009
mechanism_scores	s1p_s1pr3	LAR	pooled	10	10	-0.271	0.024	0.115
mechanism_scores	s1p_s1pr3	ISS-T	YNG	5	5	0.808	0.002	0.028
mechanism_scores	s1p_s1pr3	LAR	YNG	5	5	-0.293	0.146	0.313
mechanism_scores	s1p_s1pr3	ISS-T	OLD	5	5	0.645	0.054	0.187
mechanism_scores	s1p_s1pr3	LAR	OLD	5	5	-0.249	0.135	0.313
mechanism_scores	mmp_adam_proteolysis	ISS-T	pooled	10	10	0.535	0.011	0.086
mechanism_scores	mmp_adam_proteolysis	LAR	pooled	10	10	-0.139	0.478	0.607
mechanism_scores	mmp_adam_proteolysis	ISS-T	YNG	5	5	0.770	0.016	0.097
mechanism_scores	mmp_adam_proteolysis	LAR	YNG	5	5	-0.018	0.957	0.957
mechanism_scores	mmp_adam_proteolysis	ISS-T	OLD	5	5	0.300	0.318	0.552
mechanism_scores	mmp_adam_proteolysis	LAR	OLD	5	5	-0.261	0.315	0.552
mechanism_scores	dct_ncc_wnk_transport	ISS-T	pooled	10	10	-0.398	0.035	0.127
mechanism_scores	dct_ncc_wnk_transport	LAR	pooled	10	10	-0.149	0.461	0.606
mechanism_scores	dct_ncc_wnk_transport	ISS-T	YNG	5	5	-0.561	0.022	0.115
mechanism_scores	dct_ncc_wnk_transport	LAR	YNG	5	5	0.162	0.430	0.606
mechanism_scores	dct_ncc_wnk_transport	ISS-T	OLD	5	5	-0.235	0.431	0.606
mechanism_scores	dct_ncc_wnk_transport	LAR	OLD	5	5	-0.460	0.131	0.313
mechanism_scores	tubular_transport_broad	ISS-T	pooled	10	10	-0.267	0.280	0.528
mechanism_scores	tubular_transport_broad	LAR	pooled	10	10	-0.167	0.407	0.606
mechanism_scores	tubular_transport_broad	ISS-T	YNG	5	5	-0.246	0.456	0.606
mechanism_scores	tubular_transport_broad	LAR	YNG	5	5	0.077	0.781	0.845
mechanism_scores	tubular_transport_broad	ISS-T	OLD	5	5	-0.289	0.469	0.606
mechanism_scores	tubular_transport_broad	LAR	OLD	5	5	-0.411	0.196	0.381
mechanism_scores	oxidative_stress_nrf2	ISS-T	pooled	10	10	-0.047	0.871	0.912
mechanism_scores	oxidative_stress_nrf2	LAR	pooled	10	10	0.271	0.145	0.313
mechanism_scores	oxidative_stress_nrf2	ISS-T	YNG	5	5	-0.356	0.455	0.606
mechanism_scores	oxidative_stress_nrf2	LAR	YNG	5	5	0.425	0.166	0.343
mechanism_scores	oxidative_stress_nrf2	ISS-T	OLD	5	5	0.263	0.388	0.606
mechanism_scores	oxidative_stress_nrf2	LAR	OLD	5	5	0.118	0.641	0.756
mechanism_scores	macrophage_inflammation	ISS-T	pooled	10	10	0.380	0.072	0.216
mechanism_scores	macrophage_inflammation	LAR	pooled	10	10	-0.135	0.658	0.761
mechanism_scores	macrophage_inflammation	ISS-T	YNG	5	5	0.220	0.369	0.606
mechanism_scores	macrophage_inflammation	LAR	YNG	5	5	-0.100	0.772	0.845

Family	Feature	Arm	Age	FLT n	GC n	Effect	p	q
mechanism_scores	macrophage_inflammation	ISS-T	OLD	5	5	0.539	0.106	0.292
mechanism_scores	macrophage_inflammation	LAR	OLD	5	5	-0.170	0.741	0.829
mechanism_scores	preservation_stress_response	ISS-T	pooled	10	10	0.594	0.007	0.077
mechanism_scores	preservation_stress_response	LAR	pooled	10	10	-0.148	0.540	0.652
mechanism_scores	preservation_stress_response	ISS-T	YNG	5	5	0.385	0.295	0.540
mechanism_scores	preservation_stress_response	LAR	YNG	5	5	-0.058	0.895	0.923
mechanism_scores	preservation_stress_response	ISS-T	OLD	5	5	0.802	0.013	0.088
mechanism_scores	preservation_stress_response	LAR	OLD	5	5	-0.238	0.387	0.606
special_clock_s1p_rbm3	clock_core	ISS-T	pooled	10	10	-1.052	0.001	0.014
special_clock_s1p_rbm3	clock_core	LAR	pooled	10	10	0.118	0.420	0.573
special_clock_s1p_rbm3	clock_core	ISS-T	YNG	5	5	-1.038	0.014	0.052
special_clock_s1p_rbm3	clock_core	LAR	YNG	5	5	0.018	0.916	0.936
special_clock_s1p_rbm3	clock_core	ISS-T	OLD	5	5	-1.065	0.054	0.145
special_clock_s1p_rbm3	clock_core	LAR	OLD	5	5	0.218	0.388	0.558
special_clock_s1p_rbm3	per_cry	ISS-T	pooled	10	10	-2.020	5.96e-05	0.002
special_clock_s1p_rbm3	per_cry	LAR	pooled	10	10	0.041	0.861	0.923
special_clock_s1p_rbm3	per_cry	ISS-T	YNG	5	5	-2.001	0.004	0.029
special_clock_s1p_rbm3	per_cry	LAR	YNG	5	5	-0.142	0.658	0.789
special_clock_s1p_rbm3	per_cry	ISS-T	OLD	5	5	-2.039	0.014	0.052
special_clock_s1p_rbm3	per_cry	LAR	OLD	5	5	0.224	0.554	0.687
special_clock_s1p_rbm3	bmal_clock	ISS-T	pooled	10	10	1.680	2.84e-04	0.005
special_clock_s1p_rbm3	bmal_clock	LAR	pooled	10	10	0.079	0.747	0.854
special_clock_s1p_rbm3	bmal_clock	ISS-T	YNG	5	5	1.406	0.028	0.087
special_clock_s1p_rbm3	bmal_clock	LAR	YNG	5	5	0.504	0.138	0.283
special_clock_s1p_rbm3	bmal_clock	ISS-T	OLD	5	5	1.954	0.012	0.052
special_clock_s1p_rbm3	bmal_clock	LAR	OLD	5	5	-0.345	0.362	0.531
special_clock_s1p_rbm3	dct_wnk_expression	ISS-T	pooled	10	10	-0.450	0.066	0.164
special_clock_s1p_rbm3	dct_wnk_expression	LAR	pooled	10	10	-0.016	0.936	0.936
special_clock_s1p_rbm3	dct_wnk_expression	ISS-T	YNG	5	5	-0.750	0.014	0.052
special_clock_s1p_rbm3	dct_wnk_expression	LAR	YNG	5	5	0.228	0.325	0.517
special_clock_s1p_rbm3	dct_wnk_expression	ISS-T	OLD	5	5	-0.150	0.690	0.815
special_clock_s1p_rbm3	dct_wnk_expression	LAR	OLD	5	5	-0.260	0.356	0.531
special_clock_s1p_rbm3	aldosterone_enac_expression	ISS-T	pooled	10	10	-0.334	0.190	0.333
special_clock_s1p_rbm3	aldosterone_enac_expression	LAR	pooled	10	10	-0.418	0.148	0.288
special_clock_s1p_rbm3	aldosterone_enac_expression	ISS-T	YNG	5	5	-0.251	0.478	0.604
special_clock_s1p_rbm3	aldosterone_enac_expression	LAR	YNG	5	5	-0.068	0.806	0.879
special_clock_s1p_rbm3	aldosterone_enac_expression	ISS-T	OLD	5	5	-0.417	0.294	0.482
special_clock_s1p_rbm3	aldosterone_enac_expression	LAR	OLD	5	5	-0.769	0.146	0.288
special_clock_s1p_rbm3	s1p_axis_expression	ISS-T	pooled	10	10	0.452	0.004	0.029
special_clock_s1p_rbm3	s1p_axis_expression	LAR	pooled	10	10	-0.269	0.014	0.052
special_clock_s1p_rbm3	s1p_axis_expression	ISS-T	YNG	5	5	0.521	0.011	0.052
special_clock_s1p_rbm3	s1p_axis_expression	LAR	YNG	5	5	-0.222	0.174	0.312
special_clock_s1p_rbm3	s1p_axis_expression	ISS-T	OLD	5	5	0.383	0.100	0.231
special_clock_s1p_rbm3	s1p_axis_expression	LAR	OLD	5	5	-0.316	0.054	0.145
special_clock_s1p_rbm3	rbm3_preservation_expression	ISS-T	pooled	10	10	0.501	0.004	0.029

Family	Feature	Arm	Age	FLT n	GC n	Effect	p	q
special_clock_s1p_rbm3	rbm3_preservation_expression	LAR	pooled	10	10	-0.252	0.245	0.421
special_clock_s1p_rbm3	rbm3_preservation_expression	ISS-T	YNG	5	5	0.409	0.153	0.288
special_clock_s1p_rbm3	rbm3_preservation_expression	LAR	YNG	5	5	-0.256	0.464	0.604
special_clock_s1p_rbm3	rbm3_preservation_expression	ISS-T	OLD	5	5	0.593	0.022	0.076
special_clock_s1p_rbm3	rbm3_preservation_expression	LAR	OLD	5	5	-0.249	0.430	0.573
special_clock_s1p_rbm3	Rbm3_expression	ISS-T	pooled	10	10	-1.876	7.85e-05	0.002
special_clock_s1p_rbm3	Rbm3_expression	LAR	pooled	10	10	0.352	0.350	0.531
special_clock_s1p_rbm3	Rbm3_expression	ISS-T	YNG	5	5	-2.278	0.002	0.014
special_clock_s1p_rbm3	Rbm3_expression	LAR	YNG	5	5	0.173	0.799	0.879
special_clock_s1p_rbm3	Rbm3_expression	ISS-T	OLD	5	5	-1.475	0.038	0.113
special_clock_s1p_rbm3	Rbm3_expression	LAR	OLD	5	5	0.530	0.156	0.288
special_clock_s1p_rbm3	Per2_expression	ISS-T	pooled	10	10	-2.527	8.77e-05	0.002
special_clock_s1p_rbm3	Per2_expression	LAR	pooled	10	10	0.036	0.890	0.928
special_clock_s1p_rbm3	Per2_expression	ISS-T	YNG	5	5	-2.705	0.008	0.044
special_clock_s1p_rbm3	Per2_expression	LAR	YNG	5	5	-0.089	0.786	0.879
special_clock_s1p_rbm3	Per2_expression	ISS-T	OLD	5	5	-2.349	0.012	0.052
special_clock_s1p_rbm3	Per2_expression	LAR	OLD	5	5	0.161	0.708	0.822
special_clock_s1p_rbm3	S1pr3_expression	ISS-T	pooled	10	10	1.276	0.001	0.014
special_clock_s1p_rbm3	S1pr3_expression	LAR	pooled	10	10	-1.269	0.002	0.014
special_clock_s1p_rbm3	S1pr3_expression	ISS-T	YNG	5	5	1.572	0.008	0.044
special_clock_s1p_rbm3	S1pr3_expression	LAR	YNG	5	5	-1.401	0.028	0.087
special_clock_s1p_rbm3	S1pr3_expression	ISS-T	OLD	5	5	0.980	0.109	0.245
special_clock_s1p_rbm3	S1pr3_expression	LAR	OLD	5	5	-1.136	0.047	0.134
special_clock_s1p_rbm3	Slc12a3_expression	ISS-T	pooled	10	10	-1.114	0.060	0.154
special_clock_s1p_rbm3	Slc12a3_expression	LAR	pooled	10	10	-0.074	0.872	0.923
special_clock_s1p_rbm3	Slc12a3_expression	ISS-T	YNG	5	5	-1.373	0.079	0.189
special_clock_s1p_rbm3	Slc12a3_expression	LAR	YNG	5	5	0.069	0.926	0.936
special_clock_s1p_rbm3	Slc12a3_expression	ISS-T	OLD	5	5	-0.855	0.330	0.517
special_clock_s1p_rbm3	Slc12a3_expression	LAR	OLD	5	5	-0.218	0.644	0.786
special_clock_s1p_rbm3	Wnk4_expression	ISS-T	pooled	10	10	-0.695	0.130	0.283
special_clock_s1p_rbm3	Wnk4_expression	LAR	pooled	10	10	-0.641	0.272	0.456
special_clock_s1p_rbm3	Wnk4_expression	ISS-T	YNG	5	5	-0.761	0.137	0.283
special_clock_s1p_rbm3	Wnk4_expression	LAR	YNG	5	5	-0.631	0.475	0.604
special_clock_s1p_rbm3	Wnk4_expression	ISS-T	OLD	5	5	-0.629	0.424	0.573
special_clock_s1p_rbm3	Wnk4_expression	LAR	OLD	5	5	-0.650	0.419	0.573

Table 13: Arm-by-flight interaction tests for state, mechanism, and targeted components.

Family	Feature	ISS-T	LAR	Interaction	CI low	CI high	perm p	q
state_space	matrix_component	1.250	-0.488	1.738	0.937	2.520	0.001	0.004
state_space	dct_transport_component	-1.004	-0.377	-0.628	-1.680	0.399	0.346	0.346
state_space	immune_context_component	1.158	-0.424	1.582	0.920	2.264	5.00e-04	0.002
state_space	preservation_stress_component	1.062	-0.265	1.327	0.278	2.251	0.025	0.031
state_space	matrix_minus_dct	2.254	-0.112	2.366	0.865	3.851	0.016	0.027
mechanism_scores	ecm_organization	0.859	-0.545	1.404	0.786	2.015	0.001	0.005
mechanism_scores	fibrosis_tgfb_emt	0.068	-0.185	0.252	-0.097	0.616	0.204	0.280
mechanism_scores	tlr4_innate	0.602	-0.223	0.825	0.576	1.074	5.00e-04	0.003
mechanism_scores	integrin_cell_adhesion	0.445	-0.118	0.563	0.192	0.923	0.013	0.036
mechanism_scores	s1p_s1pr3	0.727	-0.271	0.997	0.661	1.350	5.00e-04	0.003
mechanism_scores	mmp_adam_proteolysis	0.535	-0.139	0.675	0.172	1.130	0.020	0.044
mechanism_scores	dct_ncc_wnk_transport	-0.398	-0.149	-0.249	-0.646	0.166	0.343	0.378
mechanism_scores	tubular_transport_broad	-0.267	-0.167	-0.101	-0.652	0.431	0.753	0.753
mechanism_scores	oxidative_stress_nrf2	-0.047	0.271	-0.318	-0.871	0.264	0.334	0.378
mechanism_scores	macrophage_inflammation	0.380	-0.135	0.515	-0.095	1.099	0.132	0.208
mechanism_scores	preservation_stress_response	0.594	-0.148	0.742	0.191	1.294	0.028	0.051
special_clock_s1p_rbm3	clock_core	-1.052	0.118	-1.170	-1.711	-0.704	5.00e-04	0.002
special_clock_s1p_rbm3	per_cry	-2.020	0.041	-2.061	-2.864	-1.323	1.00e-03	0.002
special_clock_s1p_rbm3	bmal_clock	1.680	0.079	1.601	0.893	2.346	0.006	0.011
special_clock_s1p_rbm3	dct_wnk_expression	-0.450	-0.016	-0.434	-0.931	0.034	0.131	0.166
special_clock_s1p_rbm3	aldosterone_enac_expression	-0.334	-0.418	0.085	-0.555	0.740	0.827	0.902
special_clock_s1p_rbm3	s1p_axis_expression	0.452	-0.269	0.721	0.442	1.001	5.00e-04	0.002
special_clock_s1p_rbm3	rbm3_preservation_expression	0.501	-0.252	0.754	0.277	1.191	0.010	0.015
special_clock_s1p_rbm3	Rbm3_expression	-1.876	0.352	-2.228	-3.131	-1.324	0.001	0.003
special_clock_s1p_rbm3	Per2_expression	-2.527	0.036	-2.563	-3.533	-1.645	1.00e-03	0.002
special_clock_s1p_rbm3	S1pr3_expression	1.276	-1.269	2.545	1.686	3.408	5.00e-04	0.002
special_clock_s1p_rbm3	Slc12a3_expression	-1.114	-0.074	-1.040	-2.266	0.072	0.138	0.166
special_clock_s1p_rbm3	Wnk4_expression	-0.695	-0.641	-0.054	-1.263	1.144	0.943	0.943

Table 14: Exploratory mechanism-switch axis projections.

Age	Arm	Axis	Simple coef.	Multi-axis coef.	Effect norm	Residual norm	Variance explained
pooled	ISS-T	matrix_high_dct_low	1.832	1.402	5.470	4.331	0.373
pooled	ISS-T	circadian_dct_state	-2.061	-1.939	5.470	4.331	0.373
pooled	ISS-T	slp_matrix	1.704	1.399	5.470	4.331	0.373
pooled	ISS-T	preservation_cold_stress	0.397	0.397	5.470	4.331	0.373
pooled	ISS-T	immune_tlr4_macrophage	1.235	1.235	5.470	4.331	0.373
pooled	ISS-T	transport_rebound	-1.527	-0.349	5.470	4.331	0.373
pooled	LAR	matrix_high_dct_low	-0.383	-0.508	1.940	1.352	0.515
pooled	LAR	circadian_dct_state	-0.009	0.307	1.940	1.352	0.515
pooled	LAR	slp_matrix	-1.025	-0.915	1.940	1.352	0.515
pooled	LAR	preservation_cold_stress	-0.213	-0.213	1.940	1.352	0.515
pooled	LAR	immune_tlr4_macrophage	-0.451	-0.451	1.940	1.352	0.515
pooled	LAR	transport_rebound	-0.617	-0.906	1.940	1.352	0.515

Table 15: Targeted clock/DCT analysis table.

Row	Family	Feature	Arm	Age	Effect	p	q	rho	corr p	corr q
effect	circadian_dct	clock_core	ISS-T	pooled	-1.052	0.001	0.013			
effect	circadian_dct	clock_core	LAR	pooled	0.118	0.420	0.627			
effect	circadian_dct	clock_core	ISS-T	YNG	-1.038	0.014	0.052			
effect	circadian_dct	clock_core	LAR	YNG	0.018	0.916	0.935			
effect	circadian_dct	clock_core	ISS-T	OLD	-1.065	0.054	0.154			
effect	circadian_dct	clock_core	LAR	OLD	0.218	0.388	0.620			
effect	circadian_dct	per_cry	ISS-T	pooled	-2.020	5.96e-05	0.002			
effect	circadian_dct	per_cry	LAR	pooled	0.041	0.861	0.922			
effect	circadian_dct	per_cry	ISS-T	YNG	-2.001	0.004	0.032			
effect	circadian_dct	per_cry	LAR	YNG	-0.142	0.658	0.831			
effect	circadian_dct	per_cry	ISS-T	OLD	-2.039	0.014	0.052			
effect	circadian_dct	per_cry	LAR	OLD	0.224	0.554	0.718			
effect	circadian_dct	bmal_clock	ISS-T	pooled	1.680	2.84e-04	0.005			
effect	circadian_dct	bmal_clock	LAR	pooled	0.079	0.747	0.874			
effect	circadian_dct	bmal_clock	ISS-T	YNG	1.406	0.028	0.089			
effect	circadian_dct	bmal_clock	LAR	YNG	0.504	0.138	0.323			
effect	circadian_dct	bmal_clock	ISS-T	OLD	1.954	0.012	0.052			
effect	circadian_dct	bmal_clock	LAR	OLD	-0.345	0.362	0.599			
effect	circadian_dct	dct_wnk_expression	ISS-T	pooled	-0.450	0.066	0.176			
effect	circadian_dct	dct_wnk_expression	LAR	pooled	-0.016	0.936	0.936			
effect	circadian_dct	dct_wnk_expression	ISS-T	YNG	-0.750	0.014	0.052			
effect	circadian_dct	dct_wnk_expression	LAR	YNG	0.228	0.325	0.599			
effect	circadian_dct	dct_wnk_expression	ISS-T	OLD	-0.150	0.690	0.849			
effect	circadian_dct	dct_wnk_expression	LAR	OLD	-0.260	0.356	0.599			
effect	circadian_dct	aldosterone_enac_expression	ISS-T	pooled	-0.334	0.190	0.396			
effect	circadian_dct	aldosterone_enac_expression	LAR	pooled	-0.418	0.148	0.323			
effect	circadian_dct	aldosterone_enac_expression	ISS-T	YNG	-0.251	0.478	0.656			
effect	circadian_dct	aldosterone_enac_expression	LAR	YNG	-0.068	0.806	0.899			
effect	circadian_dct	aldosterone_enac_expression	ISS-T	OLD	-0.417	0.294	0.565			
effect	circadian_dct	aldosterone_enac_expression	LAR	OLD	-0.769	0.146	0.323			
effect	circadian_dct	Per2_expression	ISS-T	pooled	-2.527	8.77e-05	0.002			
effect	circadian_dct	Per2_expression	LAR	pooled	0.036	0.890	0.928			
effect	circadian_dct	Per2_expression	ISS-T	YNG	-2.705	0.008	0.046			
effect	circadian_dct	Per2_expression	LAR	YNG	-0.089	0.786	0.898			
effect	circadian_dct	Per2_expression	ISS-T	OLD	-2.349	0.012	0.052			
effect	circadian_dct	Per2_expression	LAR	OLD	0.161	0.708	0.849			
effect	circadian_dct	dct_transport_component	ISS-T	pooled	-1.004	0.035	0.104			
effect	circadian_dct	dct_transport_component	LAR	pooled	-0.377	0.461	0.651			
effect	circadian_dct	dct_transport_component	ISS-T	YNG	-1.416	0.022	0.076			
effect	circadian_dct	dct_transport_component	LAR	YNG	0.408	0.430	0.627			
effect	circadian_dct	dct_transport_component	ISS-T	OLD	-0.593	0.431	0.627			
effect	circadian_dct	dct_transport_component	LAR	OLD	-1.161	0.131	0.323			
effect	circadian_dct	matrix_minus_dct	ISS-T	pooled	2.254	0.003	0.026			

Row	Family	Feature	Arm	Age	Effect	p	q	rho	corr p	corr q
effect	circadian_dct	matrix_minus_dct	LAR	pooled	-0.112	0.865	0.922			
effect	circadian_dct	matrix_minus_dct	ISS-T	YNG	3.204	0.005	0.032			
effect	circadian_dct	matrix_minus_dct	LAR	YNG	-0.782	0.361	0.599			
effect	circadian_dct	matrix_minus_dct	ISS-T	OLD	1.304	0.215	0.430			
effect	circadian_dct	matrix_minus_dct	LAR	OLD	0.558	0.533	0.711			
correlation								0.412	1.44e-04	1.44e-04
correlation								0.438	4.75e-05	7.92e-05
correlation								0.449	3.01e-05	7.52e-05
correlation								-0.547	1.53e-07	7.65e-07
correlation								-0.421	1.01e-04	1.27e-04

Table 16: Targeted S1P/S1PR3 analysis table.

Row	Family	Feature	Arm	Age	Effect	p	q	rho	corr p	corr q
effect	s1p_axis	s1p_s1pr3	ISS-T	pooled	0.727	3.89e-04	0.007			
effect	s1p_axis	s1p_s1pr3	LAR	pooled	-0.271	0.024	0.054			
effect	s1p_axis	s1p_s1pr3	ISS-T	YNG	0.808	0.002	0.012			
effect	s1p_axis	s1p_s1pr3	LAR	YNG	-0.293	0.146	0.202			
effect	s1p_axis	s1p_s1pr3	ISS-T	OLD	0.645	0.054	0.098			
effect	s1p_axis	s1p_s1pr3	LAR	OLD	-0.249	0.135	0.194			
effect	s1p_axis	s1p_axis_expression	ISS-T	pooled	0.452	0.004	0.016			
effect	s1p_axis	s1p_axis_expression	LAR	pooled	-0.269	0.014	0.037			
effect	s1p_axis	s1p_axis_expression	ISS-T	YNG	0.521	0.011	0.029			
effect	s1p_axis	s1p_axis_expression	LAR	YNG	-0.222	0.174	0.231			
effect	s1p_axis	s1p_axis_expression	ISS-T	OLD	0.383	0.100	0.163			
effect	s1p_axis	s1p_axis_expression	LAR	OLD	-0.316	0.054	0.098			
effect	s1p_axis	S1pr3_expression	ISS-T	pooled	1.276	0.001	0.012			
effect	s1p_axis	S1pr3_expression	LAR	pooled	-1.269	0.002	0.012			
effect	s1p_axis	S1pr3_expression	ISS-T	YNG	1.572	0.008	0.024			
effect	s1p_axis	S1pr3_expression	LAR	YNG	-1.401	0.028	0.059			
effect	s1p_axis	S1pr3_expression	ISS-T	OLD	0.980	0.109	0.170			
effect	s1p_axis	S1pr3_expression	LAR	OLD	-1.136	0.047	0.093			
effect	s1p_axis	matrix_component	ISS-T	pooled	1.250	0.002	0.012			
effect	s1p_axis	matrix_component	LAR	pooled	-0.488	0.120	0.181			
effect	s1p_axis	matrix_component	ISS-T	YNG	1.788	0.016	0.038			
effect	s1p_axis	matrix_component	LAR	YNG	-0.374	0.435	0.474			
effect	s1p_axis	matrix_component	ISS-T	OLD	0.712	0.095	0.162			
effect	s1p_axis	matrix_component	LAR	OLD	-0.603	0.198	0.255			
effect	s1p_axis	immune_context_component	ISS-T	pooled	1.158	8.78e-05	0.003			
effect	s1p_axis	immune_context_component	LAR	pooled	-0.424	0.229	0.275			
effect	s1p_axis	immune_context_component	ISS-T	YNG	0.930	0.002	0.012			
effect	s1p_axis	immune_context_component	LAR	YNG	-0.177	0.619	0.637			
effect	s1p_axis	immune_context_component	ISS-T	OLD	1.385	0.005	0.016			
effect	s1p_axis	immune_context_component	LAR	OLD	-0.672	0.280	0.325			
effect	s1p_axis	matrix_minus_dct	ISS-T	pooled	2.254	0.003	0.012			
effect	s1p_axis	matrix_minus_dct	LAR	pooled	-0.112	0.865	0.865			
effect	s1p_axis	matrix_minus_dct	ISS-T	YNG	3.204	0.005	0.016			
effect	s1p_axis	matrix_minus_dct	LAR	YNG	-0.782	0.361	0.406			
effect	s1p_axis	matrix_minus_dct	ISS-T	OLD	1.304	0.215	0.267			
effect	s1p_axis	matrix_minus_dct	LAR	OLD	0.558	0.533	0.565			
correlation								0.516	9.84e-07	1.31e-06
correlation								0.544	1.84e-07	5.17e-07
correlation								0.256	0.022	0.022
correlation								0.538	2.59e-07	5.17e-07

Table 17: Targeted RBM3/preservation analysis table.

Row	Family	Feature	Arm	Age	Effect	p	q	rho	corr p	corr q
effect	rbm3_preservation	Rbm3_expression	ISS-T	pooled	-1.876	7.85e-05	0.003			
effect	rbm3_preservation	Rbm3_expression	LAR	pooled	0.352	0.350	0.575			
effect	rbm3_preservation	Rbm3_expression	ISS-T	YNG	-2.278	0.002	0.029			
effect	rbm3_preservation	Rbm3_expression	LAR	YNG	0.173	0.799	0.871			
effect	rbm3_preservation	Rbm3_expression	ISS-T	OLD	-1.475	0.038	0.104			
effect	rbm3_preservation	Rbm3_expression	LAR	OLD	0.530	0.156	0.351			
effect	rbm3_preservation	rbm3_preservation_expression	ISS-T	pooled	0.501	0.004	0.033			
effect	rbm3_preservation	rbm3_preservation_expression	LAR	pooled	-0.252	0.245	0.491			
effect	rbm3_preservation	rbm3_preservation_expression	ISS-T	YNG	0.409	0.153	0.351			
effect	rbm3_preservation	rbm3_preservation_expression	LAR	YNG	-0.256	0.464	0.576			
effect	rbm3_preservation	rbm3_preservation_expression	ISS-T	OLD	0.593	0.022	0.073			
effect	rbm3_preservation	rbm3_preservation_expression	LAR	OLD	-0.249	0.430	0.575			
effect	rbm3_preservation	preservation_stress_component	ISS-T	pooled	1.062	0.007	0.038			
effect	rbm3_preservation	preservation_stress_component	LAR	pooled	-0.265	0.540	0.608			
effect	rbm3_preservation	preservation_stress_component	ISS-T	YNG	0.689	0.295	0.530			
effect	rbm3_preservation	preservation_stress_component	LAR	YNG	-0.104	0.895	0.895			
effect	rbm3_preservation	preservation_stress_component	ISS-T	OLD	1.435	0.013	0.053			
effect	rbm3_preservation	preservation_stress_component	LAR	OLD	-0.426	0.387	0.575			
effect	rbm3_preservation	preservation_stress_response	ISS-T	pooled	0.594	0.007	0.038			
effect	rbm3_preservation	preservation_stress_response	LAR	pooled	-0.148	0.540	0.608			
effect	rbm3_preservation	preservation_stress_response	ISS-T	YNG	0.385	0.295	0.530			
effect	rbm3_preservation	preservation_stress_response	LAR	YNG	-0.058	0.895	0.895			
effect	rbm3_preservation	preservation_stress_response	ISS-T	OLD	0.802	0.013	0.053			
effect	rbm3_preservation	preservation_stress_response	LAR	OLD	-0.238	0.387	0.575			
effect	rbm3_preservation	matrix_minus_dct	ISS-T	pooled	2.254	0.003	0.032			
effect	rbm3_preservation	matrix_minus_dct	LAR	pooled	-0.112	0.865	0.895			
effect	rbm3_preservation	matrix_minus_dct	ISS-T	YNG	3.204	0.005	0.033			
effect	rbm3_preservation	matrix_minus_dct	LAR	YNG	-0.782	0.361	0.575			
effect	rbm3_preservation	matrix_minus_dct	ISS-T	OLD	1.304	0.215	0.456			
effect	rbm3_preservation	matrix_minus_dct	LAR	OLD	0.558	0.533	0.608			
effect	rbm3_preservation	dct_transport_component	ISS-T	pooled	-1.004	0.035	0.104			
effect	rbm3_preservation	dct_transport_component	LAR	pooled	-0.377	0.461	0.576			
effect	rbm3_preservation	dct_transport_component	ISS-T	YNG	-1.416	0.022	0.073			
effect	rbm3_preservation	dct_transport_component	LAR	YNG	0.408	0.430	0.575			
effect	rbm3_preservation	dct_transport_component	ISS-T	OLD	-0.593	0.431	0.575			
effect	rbm3_preservation	dct_transport_component	LAR	OLD	-1.161	0.131	0.337			
correlation								-0.266	0.017	0.021
correlation								-0.266	0.017	0.021
correlation								-0.555	9.30e-08	4.65e-07
correlation								0.301	0.007	0.017
correlation								0.086	0.446	0.446

Table 18: LAR model-call table.

Evidence layer	Model A	Model B	Model C	cos	beta	Interpretation
state_space	true	false	false	-0.564	-0.199	model_A_attenuation_candidate
mechanism_scores	false	true	false	-0.725	-0.347	model_B_reversal_candidate
special_clock_slp_rbm3	true	false	false	-0.273	-0.092	model_A_attenuation_candidate
component:matrix_component	true	false	true			component_arm_divergence
component:dct_transport_component	true	false	false			component_arm_divergence
component:matrix_minus_dct	true	false	true			component_arm_divergence
component:slp_slpr3	true	true	true			component_arm_divergence
component:clock_core	true	false	true			component_arm_divergence
component:rbm3_preservation_expression	false	false	true			component_arm_divergence
mechanism_switch_axis:circadian_dct_state	true	false	true			axis_decomposition
mechanism_switch_axis:immune_tlr4_macrophage	true	true	true			axis_decomposition
mechanism_switch_axis:matrix_high_dct_low	true	true	true			axis_decomposition
mechanism_switch_axis:preservation_cold_stf	false	true	true			axis_decomposition
mechanism_switch_axis:slp_matrix	false	true	true			axis_decomposition
mechanism_switch_axis:transport_rebound	false	false	true			axis_decomposition

Table 19: Top 25 gene-priority candidates from the 2,500-gene v6 production run.

Gene	D ISS	D LAR	Emb. spec.	N ISS	N LAR	Node spec.	OSD-513	Composite	ISS —logFC—	Silent score
Rbm3	0.868	0.763	0.776	0.563	0.050	3.302	8.342	6.281	0.366	4.323
Cyp4a14	0.864	0.802	0.446	0.139	-0.095	-0.191	19.511	5.701	0.082	5.705
Bhmt	0.837	0.869	-0.255	0.560	-0.017	3.631	8.226	5.545	0.184	4.848
Per2	0.807	0.697	0.812	0.451	-0.054	2.455	7.127	5.117	0.853	-0.195
Hmgcr	0.918	0.742	1.300	0.452	0.032	2.695	4.254	5.011	0.015	5.474
Sim2	0.783	0.692	0.672	0.650	-0.008	4.360	0.250	4.886	0.420	2.557
Hac11	0.847	0.832	0.091	0.564	-0.043	3.381	5.116	4.739	0.031	5.094
Plin5	0.856	0.770	0.631	0.511	0.013	3.315	3.390	4.711	0.158	4.191
Gpt	0.876	0.781	0.702	0.491	0.046	2.836	4.489	4.622	0.077	4.658
Prodh	0.875	0.735	1.036	0.558	0.005	3.733	0.227	4.615	0.025	5.011
Ugt1a10	0.874	0.815	0.428	0.402	-0.069	1.957	8.269	4.567	0.051	4.784
Serpib8	0.829	0.876	-0.369	0.680	-0.022	4.433	2.362	4.531	0.040	4.825
Tanc2	0.844	0.821	0.158	0.603	-0.061	3.481	3.524	4.443	0.043	4.715
Syt14	0.835	0.787	0.347	0.615	0.050	3.685	2.125	4.429	0.035	4.758
Nav2	0.892	0.844	0.339	0.655	0.071	3.752	1.488	4.304	0.049	4.536
Foxo3	0.843	0.850	-0.072	0.714	0.032	4.576	-0.124	4.249	0.054	4.447
Herpud1	0.848	0.810	0.263	0.487	0.042	2.843	4.685	4.248	0.024	4.648
Tnfrsf21	0.846	0.850	-0.047	0.563	1.82e-04	3.819	2.133	4.173	0.047	4.416
Enox1	0.858	0.848	0.051	0.509	0.003	3.405	2.622	3.999	0.025	4.397
Ctxnd1	0.904	0.876	0.192	0.534	-0.044	3.159	2.920	3.980	0.160	3.447
Acmsd	0.863	0.879	-0.143	0.376	0.013	2.345	6.597	3.898	0.843	-1.344
Car4	0.857	0.782	0.541	0.681	0.156	3.062	1.773	3.898	0.178	3.237
Lpin1	0.890	0.833	0.414	0.728	-0.130	3.671	0.049	3.880	0.040	4.172
Lipa	0.920	0.822	0.721	0.508	0.055	2.858	1.789	3.879	0.129	3.559
Tnfaip2	0.833	0.750	0.606	0.540	0.028	3.372	0.215	3.821	0.349	1.982

Table 20: Top 25 gene-priority candidates from the 5,000-gene sensitivity run.

Gene	D ISS	D LAR	Emb. spec.	N ISS	N LAR	Node spec.	OSD-513	Composite	ISS —logFC—	Silent score
Rbm3	0.836	0.832	-0.026	0.572	-0.011	3.747	8.342	6.417	0.366	3.853
Bhmt	0.846	0.882	-0.409	0.568	-0.014	3.682	8.226	5.929	0.184	4.948
Kank3	0.914	0.722	1.824	0.589	0.023	3.740	0.694	5.622	0.010	6.148
Ugt1a10	0.843	0.757	0.791	0.393	-0.046	2.101	8.269	5.562	0.051	5.729
Ptpn1	0.906	0.796	1.025	0.668	-0.002	4.531	0.384	5.508	0.035	5.809
Serpinb8	0.873	0.826	0.399	0.710	-0.046	4.380	2.362	5.412	0.040	5.677
Cd200	0.863	0.776	0.799	0.507	-0.077	2.602	6.108	5.326	0.044	5.552
Per2	0.924	0.868	0.485	0.445	-0.041	2.527	7.127	5.288	0.853	-1.490
Cyp4a14	0.776	0.861	-0.888	0.120	-0.100	-0.408	19.511	5.249	0.082	5.149
Cyfp2	0.917	0.808	1.000	0.612	-0.027	3.866	1.196	5.097	0.150	4.411
Tanc2	0.820	0.828	-0.138	0.617	-0.008	4.093	3.524	4.989	0.043	5.226
Herpud1	0.834	0.813	0.152	0.517	0.025	3.201	4.685	4.787	0.024	5.184
Hacl1	0.831	0.839	-0.140	0.544	-0.032	3.332	5.116	4.775	0.031	5.116
Apex2	0.915	0.822	0.852	0.543	0.003	3.624	0.724	4.545	0.052	4.705
Ryr2	0.879	0.801	0.705	0.544	0.048	3.164	2.467	4.539	0.015	5.017
Khdc4	0.927	0.775	1.424	0.571	0.024	3.599	-0.964	4.510	0.085	4.379
Sim2	0.838	0.828	0.040	0.675	0.006	4.538	0.250	4.483	0.420	1.453
Fyb2	0.857	0.823	0.274	0.488	-0.008	3.171	3.496	4.470	0.039	4.739
Car4	0.826	0.705	1.130	0.657	0.153	2.900	1.773	4.461	0.178	3.526
Syt14	0.885	0.861	0.179	0.627	0.052	3.726	2.125	4.457	0.035	4.764
Usp2	0.873	0.858	0.087	0.617	-0.111	3.039	4.352	4.446	0.044	4.673
Safb2	0.913	0.806	0.985	0.625	-0.085	3.366	0.779	4.438	0.106	4.125
Plin5	0.817	0.782	0.285	0.471	4.58e-04	3.124	3.390	4.396	0.158	3.638
Vstm5	0.931	0.820	1.030	0.563	0.026	3.524	0.066	4.395	0.015	4.877
Atxn7l2	0.904	0.795	1.005	0.602	-0.059	3.466	0.229	4.369	0.052	4.524

Table 21: Top-k enrichment sensitivity for the gene-priority layer.

Score	Top k	Gene set	Overlap	Top n	Set n	Fold	p	q	Overlap genes
composite_score	25	ecm_organization	0	25	5	0	1	1	
composite_score	25	fibrosis_tgfb.emt	0	25	3	0	1	1	
composite_score	25	tlr4_innate	0	25	4	0	1	1	
composite_score	25	integrin_cell.adhesion	0	25	4	0	1	1	
composite_score	25	s1p.s1pr3	0	25	2	0	1	1	
composite_score	25	mmp.adam.proteolysis	0	25	4	0	1	1	
composite_score	25	dct.ncc.wnk.transport	0	25	15	0	1	1	
composite_score	25	tubular.transport.broad	0	25	4	0	1	1	
composite_score	25	oxidative_stress.nrf2	0	25	2	0	1	1	
composite_score	25	macrophage.inflammation	0	25	9	0	1	1	
composite_score	25	preservation_stress.response	0	25	10	0	1	1	
composite_score	50	ecm_organization	0	50	5	0	1	1	
composite_score	50	fibrosis_tgfb.emt	0	50	3	0	1	1	
composite_score	50	tlr4_innate	0	50	4	0	1	1	
composite_score	50	integrin_cell.adhesion	0	50	4	0	1	1	
composite_score	50	s1p.s1pr3	0	50	2	0	1	1	
composite_score	50	mmp.adam.proteolysis	0	50	4	0	1	1	
composite_score	50	dct.ncc.wnk.transport	1	50	15	3.333	0.262	1	ENSMUSG00000030340
composite_score	50	tubular.transport.broad	0	50	4	0	1	1	
composite_score	50	oxidative_stress.nrf2	0	50	2	0	1	1	
composite_score	50	macrophage.inflammation	0	50	9	0	1	1	
composite_score	50	preservation_stress.response	0	50	10	0	1	1	
composite_score	100	ecm_organization	0	100	5	0	1	1	
composite_score	100	fibrosis_tgfb.emt	0	100	3	0	1	1	
composite_score	100	tlr4_innate	0	100	4	0	1	1	
composite_score	100	integrin_cell.adhesion	0	100	4	0	1	1	
composite_score	100	s1p.s1pr3	0	100	2	0	1	1	
composite_score	100	mmp.adam.proteolysis	0	100	4	0	1	1	
composite_score	100	dct.ncc.wnk.transport	1	100	15	1.667	0.459	1	ENSMUSG00000030340
composite_score	100	tubular.transport.broad	1	100	4	6.250	0.151	1	ENSMUSG000000061742
composite_score	100	oxidative_stress.nrf2	0	100	2	0	1	1	
composite_score	100	macrophage.inflammation	0	100	9	0	1	1	
composite_score	100	preservation_stress.response	1	100	10	2.500	0.336	1	ENSMUSG000000038418
composite_score	125	ecm_organization	0	125	5	0	1	1	
composite_score	125	fibrosis_tgfb.emt	0	125	3	0	1	1	
composite_score	125	tlr4_innate	0	125	4	0	1	1	
composite_score	125	integrin_cell.adhesion	0	125	4	0	1	1	
composite_score	125	s1p.s1pr3	0	125	2	0	1	1	
composite_score	125	mmp.adam.proteolysis	0	125	4	0	1	1	
composite_score	125	dct.ncc.wnk.transport	1	125	15	1.333	0.538	1	ENSMUSG00000030340
composite_score	125	tubular.transport.broad	1	125	4	5	0.186	1	ENSMUSG000000061742
composite_score	125	oxidative_stress.nrf2	0	125	2	0	1	1	
composite_score	125	macrophage.inflammation	0	125	9	0	1	1	

Score	Top k	Gene set	Overlap	Top n	Set n	Fold	p	q	Overlap genes
composite_score	125	preservation_stress_response	1	125	10	2	0.402	1	ENSMUSG00000038418
composite_score	250	ecm_organization	0	250	5	0	1	1	
composite_score	250	fibrosis_tgfb.emt	0	250	3	0	1	1	
composite_score	250	tlr4_innate	1	250	4	2.500	0.344	1	ENSMUSG00000043496
composite_score	250	integrin_cell_adhesion	0	250	4	0	1	1	
composite_score	250	s1p.s1pr3	0	250	2	0	1	1	
composite_score	250	mmp.adam.proteolysis	0	250	4	0	1	1	
composite_score	250	dct_ncc_wnk.transport	2	250	15	1.333	0.451	1	ENSMUSG00000030340,ENSMUSG00000031766
composite_score	250	tubular_transport_broad	2	250	4	5	0.052	0.384	ENSMUSG00000061742,ENSMUSG00000063796
composite_score	250	oxidative_stress_nrf2	0	250	2	0	1	1	
composite_score	250	macrophage_inflammation	0	250	9	0	1	1	
composite_score	250	preservation_stress_response	3	250	10	3	0.070	0.384	ENSMUSG00000021250,ENSMUSG00000038418,ENSMUSG00000091971
silent_composite_score	25	ecm_organization	0	25	5	0	1	1	
silent_composite_score	25	fibrosis_tgfb.emt	0	25	3	0	1	1	
silent_composite_score	25	tlr4_innate	0	25	4	0	1	1	
silent_composite_score	25	integrin_cell_adhesion	0	25	4	0	1	1	
silent_composite_score	25	s1p.s1pr3	0	25	2	0	1	1	
silent_composite_score	25	mmp.adam.proteolysis	0	25	4	0	1	1	
silent_composite_score	25	dct_ncc_wnk.transport	0	25	15	0	1	1	
silent_composite_score	25	tubular_transport_broad	0	25	4	0	1	1	
silent_composite_score	25	oxidative_stress_nrf2	0	25	2	0	1	1	
silent_composite_score	25	macrophage_inflammation	0	25	9	0	1	1	
silent_composite_score	25	preservation_stress_response	0	25	10	0	1	1	
silent_composite_score	50	ecm_organization	0	50	5	0	1	1	
silent_composite_score	50	fibrosis_tgfb.emt	0	50	3	0	1	1	
silent_composite_score	50	tlr4_innate	0	50	4	0	1	1	
silent_composite_score	50	integrin_cell_adhesion	0	50	4	0	1	1	
silent_composite_score	50	s1p.s1pr3	0	50	2	0	1	1	
silent_composite_score	50	mmp.adam.proteolysis	0	50	4	0	1	1	
silent_composite_score	50	dct_ncc_wnk.transport	1	50	15	3.333	0.262	1	ENSMUSG00000030340
silent_composite_score	50	tubular_transport_broad	0	50	4	0	1	1	
silent_composite_score	50	oxidative_stress_nrf2	0	50	2	0	1	1	
silent_composite_score	50	macrophage_inflammation	0	50	9	0	1	1	
silent_composite_score	50	preservation_stress_response	0	50	10	0	1	1	
silent_composite_score	100	ecm_organization	0	100	5	0	1	1	
silent_composite_score	100	fibrosis_tgfb.emt	0	100	3	0	1	1	
silent_composite_score	100	tlr4_innate	0	100	4	0	1	1	
silent_composite_score	100	integrin_cell_adhesion	0	100	4	0	1	1	
silent_composite_score	100	s1p.s1pr3	0	100	2	0	1	1	
silent_composite_score	100	mmp.adam.proteolysis	0	100	4	0	1	1	
silent_composite_score	100	dct_ncc_wnk.transport	1	100	15	1.667	0.459	1	ENSMUSG00000030340
silent_composite_score	100	tubular_transport_broad	0	100	4	0	1	1	
silent_composite_score	100	oxidative_stress_nrf2	0	100	2	0	1	1	
silent_composite_score	100	macrophage_inflammation	0	100	9	0	1	1	
silent_composite_score	100	preservation_stress_response	0	100	10	0	1	1	

Score	Top k	Gene set	Overlap	Top n	Set n	Fold	p	q	Overlap genes
silent_composite_score	125	ecm_organization	0	125	5	0	1	1	
silent_composite_score	125	fibrosis_tgfb.emt	0	125	3	0	1	1	
silent_composite_score	125	tlr4_innate	0	125	4	0	1	1	
silent_composite_score	125	integrin_cell_adhesion	0	125	4	0	1	1	
silent_composite_score	125	s1p_s1pr3	0	125	2	0	1	1	
silent_composite_score	125	mmp_adam_proteolysis	0	125	4	0	1	1	
silent_composite_score	125	dct_ncc_wnk_transport	1	125	15	1.333	0.538	1	ENSMUSG00000030340
silent_composite_score	125	tubular_transport_broad	1	125	4	5	0.186	1	ENSMUSG00000061742
silent_composite_score	125	oxidative_stress_nrf2	0	125	2	0	1	1	
silent_composite_score	125	macrophage_inflammation	0	125	9	0	1	1	
silent_composite_score	125	preservation_stress_response	0	125	10	0	1	1	
silent_composite_score	250	ecm_organization	0	250	5	0	1	1	
silent_composite_score	250	fibrosis_tgfb.emt	0	250	3	0	1	1	
silent_composite_score	250	tlr4_innate	0	250	4	0	1	1	
silent_composite_score	250	integrin_cell_adhesion	0	250	4	0	1	1	
silent_composite_score	250	s1p_s1pr3	0	250	2	0	1	1	
silent_composite_score	250	mmp_adam_proteolysis	0	250	4	0	1	1	
silent_composite_score	250	dct_ncc_wnk_transport	2	250	15	1.333	0.451	1	ENSMUSG00000030340,ENSMUSG00000031766
silent_composite_score	250	tubular_transport_broad	2	250	4	5	0.052	0.574	ENSMUSG00000061742,ENSMUSG00000063796
silent_composite_score	250	oxidative_stress_nrf2	0	250	2	0	1	1	
silent_composite_score	250	macrophage_inflammation	0	250	9	0	1	1	
silent_composite_score	250	preservation_stress_response	1	250	10	1	0.652	1	ENSMUSG00000091971

Table 22: Preranked enrichment sensitivity for the gene-priority layer.

Score	Gene set	Role	Set n	Median in	Median out	p	q
composite_score	ecm_organization	outcome_anchor	5	-1.115	-0.046	0.950	0.950
composite_score	fibrosis_tgfb_emt	candidate_upstream	3	1.359	-0.047	0.143	0.430
composite_score	tlr4_innate	candidate_upstream	4	-0.861	-0.046	0.753	0.950
composite_score	integrin_cell_adhesion	candidate_upstream	4	-0.885	-0.046	0.901	0.950
composite_score	mmp_adam_proteolysis	candidate_upstream	4	-0.756	-0.046	0.920	0.950
composite_score	dct_ncc_wnk_transport	outcome_anchor	15	0.071	-0.047	0.203	0.456
composite_score	tubular_transport_broad	outcome_anchor	4	1.738	-0.047	0.008	0.076
composite_score	macrophage_inflammation	candidate_context	9	-0.400	-0.046	0.426	0.766
composite_score	preservation_stress_response	candidate_context	10	0.965	-0.047	0.052	0.233
silent_composite_score	ecm_organization	outcome_anchor	5	-3.674	0.077	0.999	0.999
silent_composite_score	fibrosis_tgfb_emt	candidate_upstream	3	-0.315	0.076	0.649	0.999
silent_composite_score	tlr4_innate	candidate_upstream	4	-1.309	0.077	0.978	0.999
silent_composite_score	integrin_cell_adhesion	candidate_upstream	4	-1.512	0.077	0.987	0.999
silent_composite_score	mmp_adam_proteolysis	candidate_upstream	4	-3.078	0.077	0.999	0.999
silent_composite_score	dct_ncc_wnk_transport	outcome_anchor	15	0.294	0.068	0.095	0.428
silent_composite_score	tubular_transport_broad	outcome_anchor	4	2.003	0.068	0.020	0.177
silent_composite_score	macrophage_inflammation	candidate_context	9	0.043	0.076	0.228	0.684
silent_composite_score	preservation_stress_response	candidate_context	10	-0.233	0.072	0.788	0.999

Table 23: External Tabula Muris Senis aging-axis projections.

Arm	Statistic	Weighted	Point	Median	CI low	CI high	p	Amplify	Dampen	Reverse	Redirect
ISS-T	beta	false	-0.003	-0.002	-0.017	0.018		0	0	0	1
ISS-T	cos	false	-0.019	-0.018	-0.108	0.106		0	0	0	1
ISS-T	rho	false	1.000	0.999	0.993	1.000		0	0	0	1
ISS-T	a_ftt_norm	false	14.550	15.978	12.856	20.835		0	0	0	1
ISS-T	a_gc_norm	false	99.992	120.558	71.142	182.477		0	0	0	1
ISS-T	residual_norm	false	14.548	15.955	12.827	20.743		0	0	0	1
LAR	beta	false	0.016	0.011	0.002	0.034		0	0	0	1
LAR	cos	false	0.175	0.123	0.033	0.230		0	0	0	1
LAR	rho	false	0.985	0.992	0.973	0.999		0	0	0	1
LAR	a_ftt_norm	false	9.236	11.097	8.210	15.176		0	0	0	1
LAR	a_gc_norm	false	99.992	120.558	70.388	182.245		0	0	0	1
LAR	residual_norm	false	9.093	10.970	8.148	15.141		0	0	0	1

Table 24: WGCNA module-level context table from the run used for v6 structural annotation.

ME	Color	Size	Flight est.	Flight q	F x Arm est.	F x Arm q	3-way est.	3-way q	Zsummary	Pres.	Top pathway	OR	Pathway q
ME17	grey60	48	0.380	6.44e-05	-0.552	4.11e-05	0.163	0.912	7.800	moderate	ecm_remodeling	29.800	0.014
ME20	royalblue	32	-0.327	9.58e-04	0.492	4.20e-04	-0.080	0.912	16.500	strong	-	-	-
ME13	salmon	103	0.297	0.001	-0.437	7.16e-04	-0.028	0.912	20.400	strong	-	-	-
ME8	pink	165	-0.298	0.003	0.301	0.042	-0.051	0.912	14.200	strong	-	-	-
ME5	green	359	0.279	0.004	-0.277	0.054	-0.048	0.912	16.500	strong	-	-	-
ME1	turquoise	841	-0.200	0.087	0.270	0.108	0.059	0.912	61.200	strong	-	-	-
ME4	yellow	474	0.218	0.087	-0.109	0.629	0.032	0.912	27	strong	-	-	-
ME9	magenta	162	-0.196	0.111	0.062	0.790	0.101	0.912	18.400	strong	-	-	-
ME16	lightcyan	79	0.192	0.151	-0.289	0.131	0.050	0.912	20.200	strong	-	-	-
ME19	lightyellow	44	0.134	0.254	-0.076	0.713	0.190	0.912	14.700	strong	-	-	-
ME6	red	297	0.132	0.254	-0.344	0.042	0.242	0.912	23.100	strong	ecm_remodeling	12.100	0.006
ME11	greenyellow	121	-0.131	0.372	0.174	0.453	-0.160	0.912	22.100	strong	-	-	-
ME15	midnightblue	80	-0.106	0.439	0.031	0.903	0.115	0.912	19.100	strong	lipid_metabolism	32.200	0.004
ME2	blue	685	0.078	0.624	-0.256	0.197	0.091	0.912	46.900	strong	oxidative_stress	15.800	0.029
ME7	black	191	-0.058	0.738	-0.059	0.790	0.162	0.912	30.800	strong	-	-	-
ME14	cyan	93	-0.046	0.738	0.003	0.979	-0.183	0.912	12.700	strong	-	-	-
ME12	tan	121	-0.037	0.855	0.096	0.713	0.002	0.992	13.600	strong	-	-	-
ME18	lightgreen	48	0.029	0.858	-0.156	0.466	0.098	0.912	11.100	strong	-	-	-
ME10	purple	147	0.021	0.885	-0.026	0.903	-0.152	0.912	23.400	strong	-	-	-
ME3	brown	554	0.012	0.906	-0.198	0.339	0.097	0.912	53.400	strong	-	-	-

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